

Tesla: Selective Label-Free Release of Canonical Adverse States Under Distribution Shift

A controlled anti-learning study with matched baselines, failure-boundary tests, and mechanism distillation

Damir Olejar

August 15, 2026

Abstract

This study asks a narrow question related to controlled anti-learning: after a scalar decision head has been deliberately placed in an adverse orientation, can future unlabeled batches be **selectively released** without consulting target labels? We call the system **Tesla**, inspired conceptually by Nikola Tesla’s valvular conduit and its direction-dependent, passive rectification.

The primary endpoint is not generic classification accuracy. For a scalar head f_θ , we measure canonical adverse mass,

$$M_D(\theta) = \Pr_{X \sim D} [f_\theta(X) \leq 0],$$

and target $M_D < 0.5$. This is equivalent to below-chance accuracy only when an external task definition independently makes $f_\theta(X) \leq 0$ the event “normal prediction is correct.” Hidden-label controls test this distinction directly.

Tesla verifies and changes only a scalar output bias, constructs conservative reserves against future movement of the scalar model score, and uses a separate input-only gate that cannot access labels, correctness, predictions, or model scores. In 130 matched trials across 13 synthetic run processes and two model classes, Tesla succeeded in 130/130 trials versus 112/130 for a one-run zero-mass bias baseline and 116/130 for a three-run baseline. Exact paired tests, with a standard family-wise-error correction for the two planned baseline comparisons, gave adjusted p -values of 1.53×10^{-5} and 1.22×10^{-4} . An independent replication produced 260/260 Tesla successes and 0/2,600 failed future holdouts.

Mechanism tests narrowed the interpretation. Rank placement alone passed 13/40 harsh matched trials; rank placement with Tesla’s dependence adjustment passed 14/40; the full Tesla method passed 40/40. However, an exploratory simplified comparator that added a reserve equal to ten times the range of the median model score observed across three runs subsequently matched the full Tesla method in 128/128 new-seed harsh trials. When equipped with the same input-only gate, this simplified comparator also rejected every tested adversarial shift no later than the first harmful point in 32/32 models. Thus the experiments do **not** establish that Tesla’s complete structural-probe ensemble is necessary.

The strongest supported result is therefore architectural: conservative one-sided adverse-state placement combined with an input-only selective-release gate can separate tested admissible and harmful covariate-shift regimes without deployment labels. External hidden-label accuracy remained at chance, ruling out a generic label-free anti-learning claim. The paper provides a complete implementation-level specification so that the experiments can be independently reimplemented without access to the original source code.

Keywords: anti-learning; selective prediction; distribution shift; covariate shift; abstention; unlabeled deployment; safety gate; adverse-state control; Tesla valve.

1. Introduction

Machine learning normally treats performance below random guessing as a failure. The literature on **anti-learning** documents cases in which a learner fits training data yet generalizes systematically worse than chance, and more recent work studies strong anti-learning regimes in which test accuracy can approach zero [1,2]. Deliberate below-chance behavior has also appeared in recent work on prompted model underperformance and sandbagging, although that problem differs substantially from the one studied here [3].

This paper investigates a different question. It does not ask whether an arbitrary hidden-label classifier can be made reliably wrong. Instead, it considers a scalar decision head with an externally designated orientation and asks:

Can an adverse state be established, frozen, and released on future unlabeled batches only when the input regime remains compatible with that state?

The distinction between **state placement** and **state release** is essential. A system that checks true accuracy after every future batch is not operating label-free; it is repeatedly recalibrating under supervision. Tesla instead permits the deployment mechanism to use only unlabeled inputs, model-side geometry, scores, and internally stored unlabeled histories. Its strictest release filter—the input-only gate—cannot evaluate the model at all.

The name **Tesla** was inspired by Nikola Tesla’s 1920 *Valvular Conduit*. Tesla’s physical conduit uses asymmetric internal geometry to impede flow more strongly in one direction than the other without conventional moving valve parts [12–14]. The analogy here is deliberately limited: the machine-learning method is not a fluid model. The design inspiration is **directionality plus passive gating**.

1.1 Terminology used throughout

To make the paper self-contained, the following terms have fixed meanings.

Term	Meaning in this paper
Tesla or full Tesla method	The selective-release algorithm evaluated in this study. “Full” distinguishes the complete method from deliberately simplified ablations and baseline comparators. The implementation was frozen before the paper-specific confirmation and challenge analyses.
Scalar model score (logit)	The single real-valued output $f_{\theta}(x)$ before thresholding at zero. “Logit” is common neural-network terminology; no probabilistic-logistic interpretation is required here.
Observed batch / observed run	One unlabeled feature matrix available while Tesla establishes its state and histories. “Run” and “batch” refer to the same experimental unit.
Future holdout batch	A newly generated batch that is not used to establish or modify the frozen state. It is used only for evaluation.
Canonical adverse mass	The fraction of examples whose scalar model score is on the designated adverse side of zero, formally $\Pr[f_{\theta}(X) \leq 0]$. This is not automatically classification accuracy.
Reserve	An additional one-sided score margin added beyond the minimum shift needed on the currently observed data, intended to protect against adverse movement on future batches.
Input-only gate	A selective-release rule that sees only feature values and previously stored feature summaries. It receives no labels, model scores, predictions, or correctness information.
Release-qualified observation	An observed batch for which all Tesla release conditions are satisfied: required observation depth, an established cross-run reserve, eligible processing geometry, and a safe shifted rank cutoff.
Post-freeze challenge suite	Eight synthetic distribution families created only after the Tesla implementation had been frozen. “Post-freeze” means the algorithm was not changed in response to these distributions; it does not mean the experiment was preregistered.
Ten-times median-range reserve comparator	A deliberately simpler baseline designed after the primary experiment had already been examined. After three observed runs it adds a reserve equal to ten times the range of the three pre-shift median model scores. Because it was designed post hoc, it is treated as exploratory rather than as a preplanned confirmatory baseline.

Term	Meaning in this paper
Matched trial	A comparison in which Tesla and the baseline begin from the same initial model and receive the same generated observed and future batches. This removes random differences in initialization or data stream from the method comparison.
Ablation	A deliberately simplified version of the method in which one or more components are removed. Ablations are used to test which mechanisms are actually contributing to the result.
Covariate shift	A change in the distribution of the input features presented to a frozen model. In this paper the term refers to input-distribution change; it does not imply that hidden target labels are available.

1.2 Canonical orientation is necessary

Let $\hat{Y}(X)$ be any binary classifier and let the hidden target be Y . From unlabeled X and model outputs alone, classification accuracy is not identifiable in general because replacing the unknown target by $1 - Y$ leaves all unlabeled observables unchanged while complementing accuracy.

Therefore any label-free below-chance claim needs an independently justified target orientation.

The synthetic experiments in this paper do **not** manufacture that missing information. They study a more primitive quantity: the fraction of model scores on a designated side of zero. We call this **canonical adverse mass**. If a future application has independently established that this side corresponds to normal correctness, then canonical adverse mass can be interpreted as normal accuracy. Otherwise it remains an output-side statistic.

This scope boundary is not merely verbal. Section 7 reports hidden external-label audits that remain at chance even when canonical adverse mass is driven to zero.

1.3 What the Tesla method actually does

Tesla is not a conventional learning algorithm during the intervention studied here. It receives an already instantiated model. It then:

1. verifies that a trainable scalar parameter causes a uniform shift of the final score;
2. computes a conservative target below the 0.5 mass boundary;
3. estimates reserves for possible future adverse score displacement;
4. changes only that verified scalar bias;
5. stores unlabeled input histories;
6. releases only after a separate admissibility criterion is satisfied.

No optimizer step, feature-weight update, label lookup, replay update, or supervised calibration occurs inside the Tesla intervention evaluated here.

Figure~1 summarizes the operational separation between input-only admissibility, score-side reserve construction, scalar state placement, and the final release decision.

1.4 Research questions

The experiments answer five progressively harder questions.

RQ1 — Persistence. Does an adverse state established on observed unlabeled runs remain below 0.5 on fresh future batches?

RQ2 — Baselines. Is performance better than simply moving the bias far enough to make one or three observed batches have zero canonical mass?

RQ3 — Mechanism. Which parts of the reserve machinery matter, and can a much simpler conservative reserve match the full system?

RQ4 — Release safety. When a future distribution is deliberately moved until the frozen state fails, does the input-only gate reject before or at the first harmful point?

RQ5 — Scope. Does canonical adverse mass imply below-chance accuracy against arbitrary hidden labels?

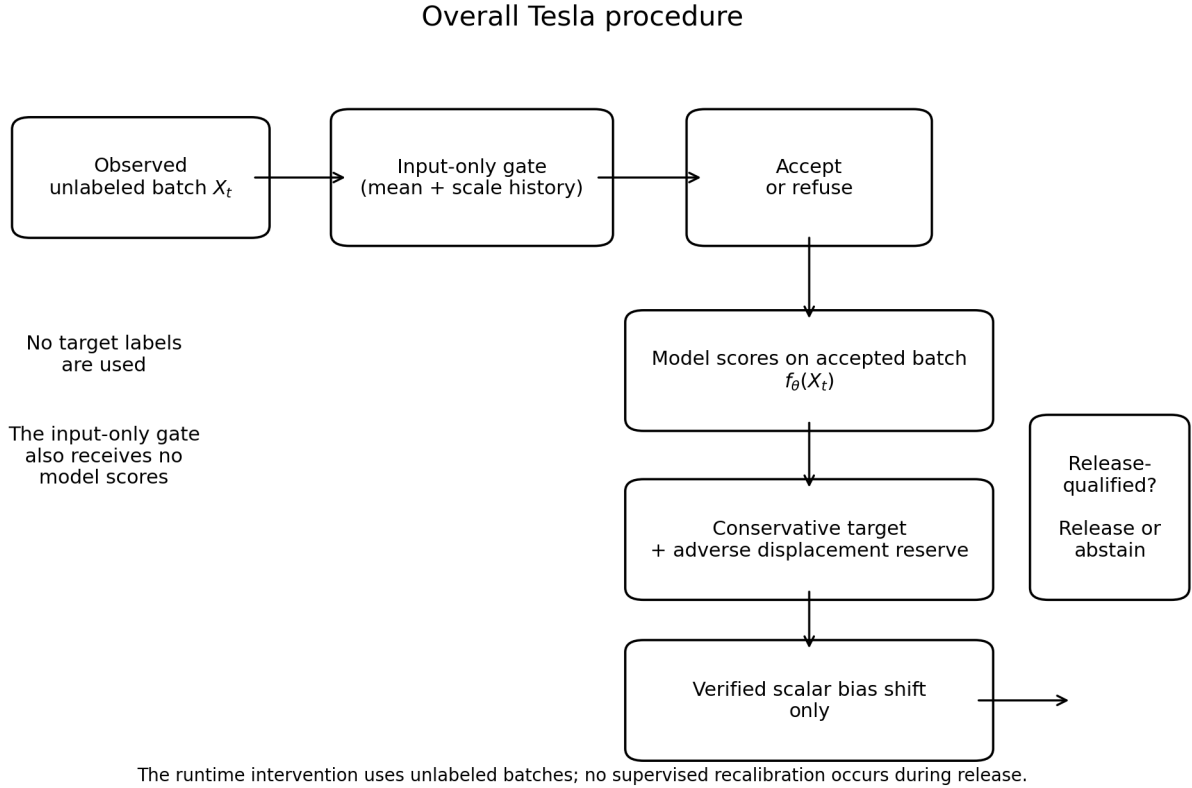


Figure 1. Overall Tesla selective-release procedure. The input-only gate is evaluated from unlabeled input summaries and has no access to target labels or model scores. Accepted batches are then used to construct a conservative target and adverse displacement reserve before a verified scalar bias is shifted.

1.5 Contributions

The main contributions are:

- a precise formulation of **selective label-free release for a canonical adverse state**;
- a matched experimental protocol using identical initial models and identical stochastic streams across methods;
- strong minimal-bias controls and an exploratory stronger conservative-reserve control;
- a targeted adversarial failure path in which failure is intentionally induced rather than merely searched for;
- an input-only gate whose complete formula is specified in the manuscript;
- explicit hidden-label negative controls;
- an ablation that separates rank placement, dependence discounting, reserve construction, and gating;
- a complete paper-level operational specification, including model initialization, random generators, seed schedules, constants, stopping rules, statistical formulas, and probe construction, so that no source-code access is required.

1.6 Claim ledger

Claim	Status
A positive uniform score shift cannot increase empirical canonical mass	Proved
The full Tesla method persists below 0.5 on the tested synthetic future-run processes	Supported empirically
The full Tesla method outperforms minimal one- and three-run zero-mass baselines on matched trials	Supported empirically

Claim	Status
Rank placement alone is insufficient on the tested coherent-shift processes	Supported empirically
Tesla’s complete structural-probe ensemble is necessary	Not supported
A simpler empirical between-run reserve can match the full Tesla method on the tested harsh suite	Supported exploratorily and on new seeds
The input-only gate rejects no later than failure on the tested adversarial path	Supported for both the full Tesla method and the strong simple-reserve comparator
Canonical adverse mass implies below-chance arbitrary-label accuracy	Falsified / not supported
The internal nominal 10^{-24} rank risk is a distribution-free deployment guarantee	Not claimed
Tesla provides a universal theorem against arbitrary distribution shift	Not claimed

2. Related Work

2.1 Anti-learning

Kowalczyk and Chapelle analyzed the anti-learning phenomenon for a class-symmetric polyhedron and distinguished it from ordinary overfitting [1]. Lisitsa and Vernitski later formalized **strong anti-learning**, including parity problems for which increasing the training sample can worsen test accuracy toward zero [2].

Those works study below-chance **true classification accuracy**. The present manuscript instead studies a controlled canonical state and its unlabeled release. The terms should not be conflated unless an application supplies the missing correctness orientation.

Cacioli’s 2026 preprint on prompted underperformance in small language models is useful as a contemporary contrast: generic sandbagging prompts often produced positional biases rather than answer-aware below-chance performance, while an explicit anti-task could push some models below chance [3]. That work studies behavioral underperformance, not selective release under covariate shift.

2.2 Estimating performance without labels

Methods for estimating classifier risk from unlabeled data already exist under structural assumptions. Donmez, Lebanon, and Balasubramanian developed unsupervised estimators of classification and regression error [4]. Jaffe, Nadler, and Kluger estimated multiple classifier accuracies from unlabeled predictions under conditional-independence assumptions [5]. Margin risk can also be estimated without labels when additional information such as the marginal label distribution is available [6].

Tesla does not estimate arbitrary hidden-label accuracy. The hidden-label negative control is included specifically to prevent that interpretation.

2.3 Covariate shift, abstention, and testable deployment

Selective classification under covariate shift allows a system to reject examples that are insufficiently supported by the training distribution. Kalai and Kanade developed efficient learning with arbitrary covariate shift in a rejection-enabled setting [7].

Klivans, Stavropoulos, and Vasilyan formalized **Testable Learning with Distribution Shift**, in which labeled training data and unlabeled test data are accompanied by a test that can reject deployment when the required guarantee cannot be certified [8]. The **Suitability Filter** similarly uses unlabeled deployment signals to detect performance degradation [9]. Timans et al. developed continuous sequential monitoring for risk violations under unknown shift [10], while recent work continues to improve provable algorithms for covariate-shift estimation [11].

Tesla is conceptually closest to this **test-before-release** family. The unusual sign of its target—an intentionally adverse canonical state—does not eliminate the underlying deployment problem: labels are absent, so the system

needs a criterion for whether a new input distribution is still admissible.

2.4 Tesla-valve inspiration

Nikola Tesla’s *Valvular Conduit*, U.S. Patent 1,329,559, was patented February 3, 1920 from an original February 21, 1916 application [12]. The conduit permits easier flow in one direction and uses rigid internal geometry to oppose reverse flow. Modern experiments have quantified this direction-dependent fluidic resistance and its nonlinear flow regime [13,14].

The present method borrows only the systems metaphor: **rectified orientation and passive structural gating**.

3. Formal Problem Statement

3.1 Canonical adverse mass

Let

$$f_\theta : \mathcal{X} \rightarrow \mathbb{R}$$

be a scalar head.

For input distribution D ,

$$M_D(\theta) = \Pr_{X \sim D} [f_\theta(X) \leq 0].$$

For a finite batch $X_1, \dots, X_n$,

$$\widehat{M}_n(\theta) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[f_\theta(X_i) \leq 0].$$

The controlled adverse regime is

$$\widehat{M}_n(\theta) < \tau, \quad \tau = 0.5.$$

The phrase **canonical anti-learning** is justified only if an external task definition establishes

$$\mathbf{1}[f_\theta(X) \leq 0] = \mathbf{1}[\text{normal prediction is correct}].$$

Otherwise the paper uses **canonical adverse mass**.

Figure~2 gives a schematic view of the canonical adverse-mass endpoint and the one-sided scalar intervention.

3.2 Proposition 1: monotonicity of scalar placement

Let $z_i = f_\theta(X_i)$. Apply a uniform score shift

$$z_i(\Delta) = z_i + \Delta, \quad \Delta \geq 0.$$

Then

$$\widehat{M}_n(\Delta) = \frac{1}{n} \sum_i \mathbf{1}[z_i + \Delta \leq 0]$$

is non-increasing in Δ .

Proof. If $\Delta_2 \geq \Delta_1$, then

$$\{i : z_i + \Delta_2 \leq 0\} \subseteq \{i : z_i + \Delta_1 \leq 0\}.$$

Therefore the count cannot increase. $\square$

This proposition is important because it establishes what is *not* scientifically difficult: making an observed batch adverse is trivial if a verified uniform scalar bias exists. The difficult question is future persistence and label-free release.

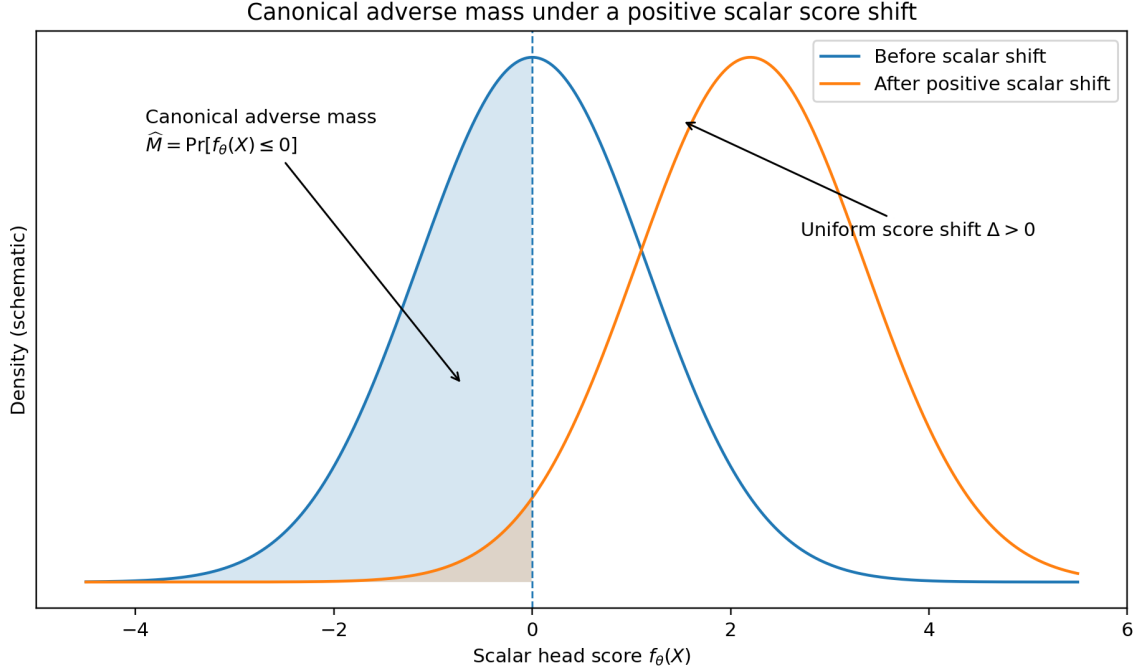


Figure 2. Schematic of canonical adverse mass under a positive uniform scalar score shift. The shaded area to the left of zero is the empirical or population mass counted as adverse by the canonical head. The figure is conceptual rather than a fitted density from an experiment.

3.3 Proposition 2: arbitrary hidden-label orientation is not identifiable

For any prediction $\hat{Y}$ and target Y , define $Y' = 1 - Y$. Then

$$\Pr[\hat{Y} = Y] + \Pr[\hat{Y} = Y'] = 1.$$

The unlabeled distribution of X and the model output is unchanged by replacing the hidden target with its complement. Hence, without additional assumptions, unlabeled observables cannot identify whether arbitrary classification accuracy is above or below 0.5.

3.4 Selective release

For a future batch X , let

$$G(X) \in \{0, 1\}$$

be the input-only gate, where $G = 1$ means accept.

Define harm as

$$H(X) = \mathbf{1}[\widehat{M}(X) \geq 0.5].$$

Relevant deployment quantities are:

$$\text{coverage} = \Pr[G(X) = 1],$$

$$\text{harmful-acceptance risk} = \Pr[H(X) = 1 \wedge G(X) = 1],$$

and, along an ordered stress path indexed by δ ,

$$\delta_{\text{reject}} = \min\{\delta : G(X_\delta) = 0\},$$

$$\delta_{\text{fail}} = \min\{\delta : \widehat{M}(X_\delta) \geq 0.5\}.$$

The tested safety-precedence event is

$$\delta_{\text{reject}} \leq \delta_{\text{fail}}.$$

3.5 Exchangeability-based rank target

One component of the Tesla method uses an order-statistic bound. It should be interpreted carefully.

Let n be observed rows and m future rows. Suppose the observed and future rows are exchangeable under a common row distribution. If at most k observed rows are allowed below the selected cutoff and future failure means at least

$$t = \left\lceil \frac{m}{2} \right\rceil$$

future rows fall below the same cutoff, then in the combined ordering of $n+m$ rows the observed/future membership labels are exchangeable.

The Tesla method does not evaluate an exact hypergeometric tail. For each candidate k , set

$$L = k + t, \quad p = \frac{n}{n+m}, \quad a = \frac{k}{L}, \quad \epsilon = p - a.$$

When $\epsilon > 0$, two log upper bounds are computed:

$$B_{\text{Serf}} = -2L\epsilon^2 \frac{n+m}{n+m-L+1},$$

and

$$B_{\text{KL}} = -L D_{\text{KL}}(\text{Bern}(a) \parallel \text{Bern}(p)).$$

where D_{KL} is the Kullback–Leibler divergence between two Bernoulli distributions. The first expression is a Serfling-style finite-population concentration bound for sampling without replacement [15].

The implemented log bound is

$$B(k) = \min(B_{\text{Serf}}, B_{\text{KL}}).$$

The Tesla method chooses the largest

$$k \leq \lfloor 0.5n \rfloor$$

such that

$$B(k) \leq \log(10^{-24}).$$

The corresponding nominal rank target is

$$\tau_{\text{rank}} = k/n.$$

The value 10^{-24} is an **internal nominal risk budget under the exchangeability construction**, not a demonstrated 10^{-24} probability of failure for the non-IID shift processes used in this paper. The non-IID experiments are empirical stress tests, and the paper makes no distribution-free 10^{-24} guarantee.

4. Tesla Method: Complete Operational Specification

This section is written so that the runtime can be independently reimplemented without access to the original source code. Section 11 and the Supplementary Methods give all numerical constants and deterministic construction details.

4.1 State variables

For each model instance, maintain:

- cumulative uniform score shift s_{cum} , initialized to 0;
- accepted-run count t , initialized to 0;
- required release depth r , initialized to 2;
- up to eight full-run feature-mean vectors;
- up to eight full-run feature log-scale vectors;
- up to sixteen absolute run-location values;
- up to eight feature-drift mean summaries;
- up to eight global and diagonal feature-scale summaries;
- up to eight covariance-projection summaries;
- up to eight leading covariance eigenvalue summaries.

All histories are first-in/first-out once their maximum length is exceeded.

4.2 Verifying the scalar output bias

Tesla may intervene only if it can verify a trainable scalar parameter b whose perturbation uniformly shifts the final scalar score.

Candidate parameters must:

- require gradients;
- contain exactly one scalar;
- have a parameter name ending in `bias`.

Use the first

$$r_b = \min(64, n)$$

rows as a probe. Let the original scores be z .

For each candidate, temporarily perturb the parameter by

$$+\epsilon, \quad -\epsilon, \quad +16\epsilon, \quad \epsilon = 0.001.$$

Let the resulting score differences be d_+ , d_- , and d_{16} . Estimate the per-parameter score gain

$$g = \frac{\text{mean}(d_+)}{\epsilon}.$$

Also estimate

$$g_- = \frac{\text{mean}(d_-)}{-\epsilon}, \quad g_{16} = \frac{\text{mean}(d_{16})}{16\epsilon}.$$

Accept the parameter only if:

1. $|g| > 10^{-8}$;
2. every tested score difference is finite;
3. the largest row-wise deviation from its corresponding mean shift is no greater than

$$\max(10^{-4}, |\text{mean}(d_+)|10^{-4});$$

4. the relative gain discrepancy

$$\frac{\max(|g_- - g|, |g_{16} - g|)}{\max(|g|, 10^{-12})} \leq 0.002.$$

Once verified, cache the parameter identity and gain g .

A desired score shift Δ is implemented as a parameter change

$$\Delta b = \Delta/g.$$

4.3 Input-only full-run summary

For every accepted observed run $X \in \mathbb{R}^{n \times d}$, compute from **all rows**, not from a model-selected subset,

$$\begin{aligned}\mu_j &= \frac{1}{n} \sum_{i=1}^n X_{ij}, \\ v_j &= \max \left(10^{-24}, \frac{1}{n} \sum_i X_{ij}^2 - \mu_j^2 \right), \\ \ell_j &= \frac{1}{2} \log v_j.\end{aligned}$$

Store $\mu = (\mu_1, \dots, \mu_d)$ and $\ell = (\ell_1, \dots, \ell_d)$ in histories of maximum length 8.

Non-finite input values cause refusal.

4.4 Input-only gate

The gate is evaluated **before any model forward pass** on a future run once at least two historical summaries exist. Before two historical summaries exist, the run is accepted with reason `BUILDING_INPUT_HISTORY`.

Let prior mean-history matrix be

$$M \in \mathbb{R}^{h \times d},$$

and prior log-scale history be

$$L \in \mathbb{R}^{h \times d},$$

where $h \leq 8$.

Use

$$d_2(h) \in \{1.128, 1.693, 2.059, 2.326, 2.534, 2.704, 2.847\}$$

for $h = 2, \dots, 8$.

For each feature,

$$c_j = \text{median}_r M_{rj},$$

$$w_j = \exp(\text{median}_r L_{rj}),$$

$$b_j = \frac{\max_r M_{rj} - \min_r M_{rj}}{d_2(h)}.$$

Define multivariate historical radial displacement

$$\rho_r = \sqrt{\frac{1}{d} \sum_j (M_{rj} - c_j)^2},$$

$$\rho_{\max} = \max_r \rho_r,$$

and

$$\rho_{\text{between}} = 32\rho_{\max}.$$

For the current run, using its full-run sample size n ,

$$s_j^{(\mu)} = \sqrt{b_j^2 + \rho_{\text{between}}^2 + \left(\frac{w_j}{\sqrt{n}}\right)^2 + (1.5w_j)^2}.$$

The standardized mean displacement is

$$z_j^{(\mu)} = \frac{\mu_j^{\text{current}} - c_j}{\max(s_j^{(\mu)}, 10^{-12})}.$$

Here **RMS** means root mean square. Compute

$$z_{\text{mean,RMS}} = \sqrt{\frac{1}{d} \sum_j (z_j^{(\mu)})^2},$$

$$z_{\text{mean,max}} = \max_j |z_j^{(\mu)}|.$$

For log scales,

$$c_j^{(\ell)} = \text{median}_r L_{rj},$$

$$b_j^{(\ell)} = \frac{\max_r L_{rj} - \min_r L_{rj}}{d_2(h)},$$

$$s_j^{(\ell)} = \sqrt{(b_j^{(\ell)})^2 + 0.25^2},$$

$$z_j^{(\ell)} = \frac{\ell_j^{\text{current}} - c_j^{(\ell)}}{\max(s_j^{(\ell)}, 10^{-12})}.$$

Accept only if all four inequalities hold:

$$z_{\text{mean,RMS}} \leq 4.5,$$

$$z_{\text{mean,max}} \leq 10,$$

$$z_{\text{scale,RMS}} \leq 5,$$

$$z_{\text{scale,max}} \leq 10.$$

The gate does not receive the model, scores, predictions, labels, correctness, or canonical mass.

The gate architecture is summarized in Figure~3.

4.5 Input-only serial-dependence score and model-row budget

After a run passes the gate, record its full-run summary and estimate whether the run contains strong row-wise block dependence.

For $n \geq 64$, split the run into

$$B = \min(32, \max(4, \lfloor n/16 \rfloor))$$

contiguous blocks with approximately equal length.

For block b , feature j , let $\bar{X}_{bj}$ be the block mean, μ_j the full-run mean, v_j the full-run variance, and m_b the block length. Define

$$R_{bj} = \frac{(\bar{X}_{bj} - \mu_j)^2}{v_j/m_b}.$$

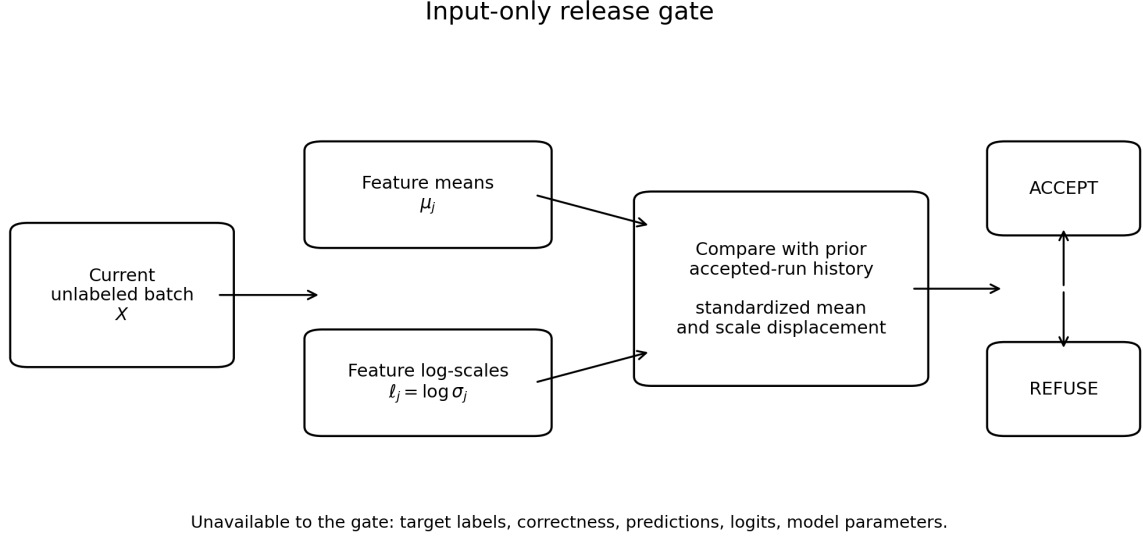


Figure 3. Input-only release gate. The gate compares current feature means and log-scales with histories from previously accepted runs. Target labels, correctness, predictions, logits, and model parameters are unavailable to this gate.

For each block,

$$q_b = \text{median}_j R_{bj}.$$

The input-only serial design score is

$$D_{\text{serial}} = \max(1, Q_{0.75}(q_1, \dots, q_B)).$$

If

$$D_{\text{serial}} \geq 2.5,$$

the model-evaluation sample cap is 1,024 rows; otherwise it is 320.

4.6 Deterministic model-row sampling

If n is no larger than the current cap, use all rows.

Otherwise let the contiguous window length be 16 and

$$W = \left\lfloor \frac{\text{cap}}{16} \right\rfloor.$$

Divide the row range into W equal-proportion strata. In each stratum choose one contiguous 16-row window. The start position is determined only by:

- number of rows;
- observation index;
- stratum index;

through the 64-bit mixing function given in Supplement S2. No feature value, score, or label enters the row choice.

4.7 Sample-to-full representativeness check

When a sampled model-evaluation set is used, compare its feature mean and log scale to the full-run input-only summary.

For each feature, normalize the sample/full mean difference by the full-run scale. Accept sample geometry only if:

$$\text{RMS}(z_\mu) \leq 0.35,$$

$$\max |z_\mu| \leq 1.25,$$

$$\text{RMS}(\Delta \log \sigma) \leq 0.35,$$

$$\max |\Delta \log \sigma| \leq 1.25.$$

At observation 2, a sampled run can be release-eligible only if this representativeness certificate passes. Observations

$$3, 4, 5, 6, 8, 10, 12$$

are full-processing milestones, and all observations after 12 are processed fully.

4.8 Conservative target mass

Let n_p be the placement-row count and m the expected future-row count. In these experiments m equals the run size.

First calculate the exchangeability rank target τ_{rank} from Section 3.5.

Also calculate standard-error cushions, where **SE** denotes standard error:

$$SE_o = \sqrt{0.25/n_p}, \quad SE_f = \sqrt{0.25/m}.$$

Define

$$c_o = \min(0.22, \max(0.005, 10SE_o)),$$

$$c_f = \min(0.22, \max(0.005, 10SE_f)).$$

The preliminary heuristic target is

$$\tau_h^{(0)} = \max(0, 0.5 - \max(c_o, c_f)).$$

The preliminary target is

$$\tau_0 = \min(\tau_h^{(0)}, \tau_{\text{rank}}).$$

4.9 Score-dependence design effect

Let the preliminary score cutoff be the order statistic at index

$$k_0 = \lfloor n_a \tau_0 \rfloor,$$

where n_a is the analysis-row count.

Define

$$I_i = \mathbf{1}[z_i \leq c_0], \quad p = \frac{1}{n_a} \sum_i I_i.$$

Center

$$u_i = I_i - p.$$

For lags $l = 1, \dots, \min(8, n_a - 2)$,

$$r_l = \frac{\sum_{i=1}^{n_a-l} u_i u_{i+l}}{\sum_i u_i^2}.$$

Use only positive correlation beyond 0.02:

$$\rho_l = \max(0, r_l - 0.02).$$

Taper by

$$w_l = 1 - \frac{l}{L+1},$$

where L is the maximum lag.

Then

$$D_{\text{AC}} = 1 + 2 \sum_l w_l \rho_l.$$

For slow block dependence, use block size

$$b = \min(256, \max(16, \lfloor \sqrt{n_a} \rfloor)).$$

If at least four complete blocks exist and p is non-degenerate, compare observed block-mean variance to the IID Bernoulli variance:

$$D_{\text{raw}} = \frac{\text{Var}(\bar{I}_{\text{block}})}{p(1-p)/b},$$

$$D_{\text{block}} = \max(1, 1 + 0.75(D_{\text{raw}} - 1)).$$

The final design effect is

$$D = \min(8, \max(1.15, D_{\text{AC}}, D_{\text{block}})).$$

Use effective sample sizes

$$n_{\text{eff}} = \max(1, \lfloor n_p/D \rfloor),$$

$$m_{\text{eff}} = \max(1, \lfloor m/D \rfloor).$$

Repeat the heuristic and rank-target calculations using n_{eff} and m_{eff} . The final target mass is

$$\tau_* = \min(\tau_0, \tau_h^{(\text{eff})}, \tau_{\text{rank}}^{(\text{eff})}, 0.47).$$

The 0.47 term is the configured 0.03 noise-floor cushion below 0.5.

4.10 Target order statistic and two-row guard

Let the placement logits be $z_1, \dots, z_{n_p}$.

Set

$$k_* = \max(0, \min(n_p - 1, \lfloor \tau_* n_p \rfloor - 2)).$$

Let

$$c_* = \text{orderstat}_{k_*}(z).$$

This is the pre-shift target-rank cutoff.

4.11 Between-run location history

Let $c_{0.5}$ be the median score cutoff on the current analysis run and let s_{cum} be the cumulative score shift already applied by Tesla.

Define the absolute run-location statistic

$$r_t = s_{\text{cum}} - c_{0.5}.$$

Store at most 16 values.

For $h \geq 2$, estimate a scale from the observed range:

$$\sigma_r = \frac{\max r - \min r}{d_2(h)},$$

using the same d_2 constants through $h = 16$.

Define:

$$R_{\text{sym}} = 10\sigma_r,$$

$$R_{\text{adverse}} = \max(0, \max r + 4\sigma_r - r_t),$$

$$R_{\text{center}} = \max(0, \text{median}(r) + 10\sigma_r - r_t).$$

The history reserve is

$$R_{\text{hist}} = \max(R_{\text{sym}}, R_{\text{adverse}}, R_{\text{center}}).$$

4.12 Tail-establishment criterion

For each history prefix of length at least 2, sort the prefix and take at most the four largest values. Let s_{top} be the maximum adjacent spacing among those top values.

Define prefix upper bound

$$U_h = \max r_{1:h} + 4s_{\text{top}}.$$

At the current history length, compute the most recent positive growth

$$g_h = \max(0, U_h - U_{h-1}).$$

Tail stability holds if

$$g_h \leq 0.2\sigma_r.$$

The tail is considered established after at least two history values and one consecutive stable update.

4.13 Robust logit scale and base reserve

Let q_{25} and q_{75} be the score quartiles on the analysis rows. Define

$$s_z = \max\left(10^{-12}, \frac{q_{75} - q_{25}}{1.3489795003921634}\right).$$

The base score reserve is

$$R_{\text{base}} = \left[1.1 \min\left(1, \sqrt{256/m_{\text{eff}}}\right)\right] s_z.$$

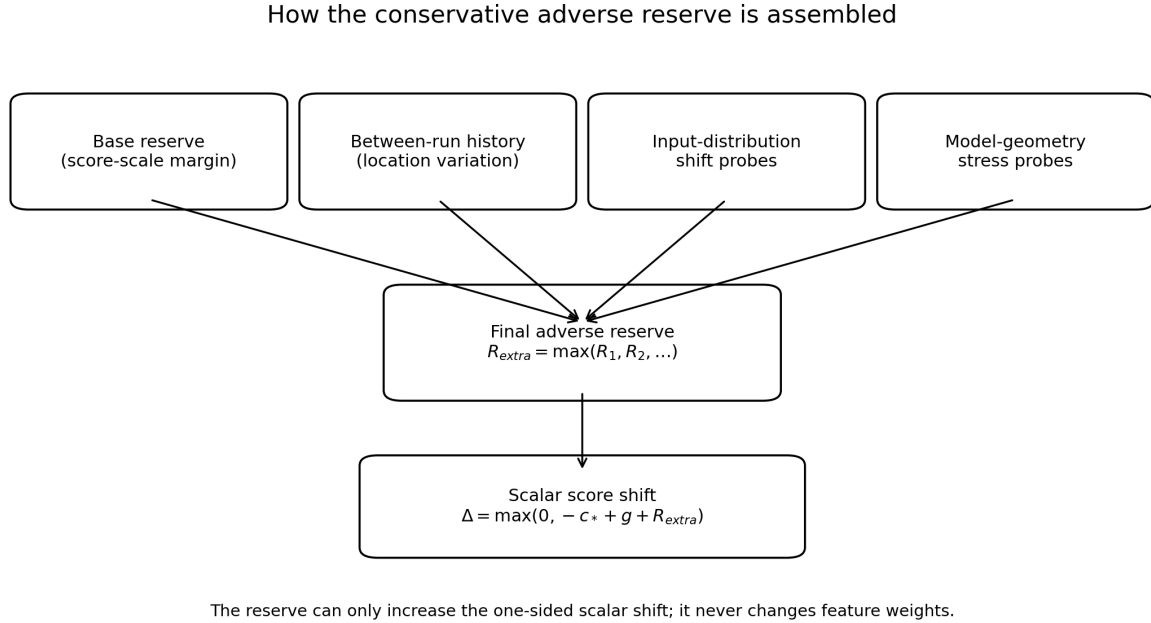


Figure 4. Assembly of the conservative adverse displacement reserve. The full Tesla implementation computes several nonnegative reserve terms and applies their maximum as additional scalar protection beyond the target-rank cutoff.

4.14 Structural reserves

Tesla computes a set of additional nonnegative score margins:

$$R_{\text{drift}}, R_{\text{cov}}, R_{\text{gate}}, R_{\text{support}}, R_{\text{marginal}}, R_{\text{shape}}, R_{\text{direction}}, R_{\text{pair}}, R_{\text{opaque}}, R_{\text{sphere}}, R_{\text{affine}}.$$

Every probe is label-free, but not every probe is input-only. Structural probes may use model scores, model gradients, or model parameters **after their synthetic or actual-support probe points are fixed**. They may change only the scalar reserve. They do not choose real training labels or externally scored outcomes.

The complete constructions are specified in Supplement S3.

The final extra score reserve is

$$R_{\text{extra}} = \max(R_{\text{base}}, R_{\text{hist}}, R_{\text{drift}}, R_{\text{cov}}, R_{\text{gate}}, R_{\text{support}}, R_{\text{marginal}}, R_{\text{shape}}, R_{\text{direction}}, R_{\text{pair}}, R_{\text{opaque}}, R_{\text{sphere}}, R_{\text{affine}}).$$

Figure~4 shows how the full method reduces multiple conservative mechanisms to a single one-sided score reserve.

4.15 Cross-run establishment

Cross-run protection is considered established if at least one of the following holds:

1. the historical upper-tail criterion is established;
2. the gate-envelope reserve is positive and at least R_{hist} ;
3. a structural envelope reserve is positive and at least R_{hist} , where

$$R_{\text{structural}} = \max(R_{\text{affine}}, R_{\text{opaque}}, R_{\text{sphere}}).$$

4.16 Scalar placement

Define numerical guard

$$g = \max(10^{-6}, 10^{-6}|c_*|).$$

The requested score shift is

$$\Delta = \max(0, -c_* + g + R_{\text{extra}}).$$

Apply Δ only through the verified scalar bias.

Because the bias linearity was verified, post-shift placement scores are computed as

$$z'_i = z_i + \Delta.$$

A second direct placement is allowed only as a numerical/tie guard if the predicted post-shift mass remains above τ_* .

The intervention must never increase canonical mass beyond a one-row numerical tolerance. A violation is treated as an error rather than silently accepted.

4.17 Adaptive release depth

Start with required observation depth

$$r = 2.$$

If cross-run protection is not established when $t \geq 2$, increase the requirement to the next even observation:

$$r = 2 \left\lceil \frac{t+1}{2} \right\rceil,$$

capped at 12.

If the nominal rank bound is not certifiable at the effective sample size, force the requirement to 12 and drive the observed target toward zero.

4.18 Final release condition

Let the total score shift in the current intervention be Δ_{total} .

Define shifted target-rank cutoff

$$c_{\text{shifted}} = c_* + \Delta_{\text{total}}.$$

Rank release is safe only if

$$c_{\text{shifted}} > 5 \times 10^{-7}.$$

An observation is **release-qualified** only if:

- release processing is eligible;
- $t \geq r$;
- cross-run protection is established;
- the shifted target-rank condition passes.

In the primary 5,000-row matched experiment, all 130 Tesla trials reached release at exactly the third observed batch.

4.19 Runtime pseudocode

```
STATE:
    t = 0                                # accepted observation count
    required_depth = 2
    cumulative_score_shift = 0
    input_mean_history = []
```

```

input_logscale_history = []
run_location_history = []
probe_histories = ...

FOR EACH OBSERVED UNLABELED RUN X:

  if t >= 2:
    if INPUT_ONLY_GATE(X, previous input histories) == REFUSE:
      return REFUSE_RUN

  record full-input mean and log-scale history from all rows of X
  t <- t + 1

  estimate input-only serial design score
  cap <- 1024 if serial score >= 2.5 else 320
  select deterministic model-evaluation rows using only (n, t, cap)

  if t is 3,4,5,6,8,10,12 or t > 12:
    placement rows <- all rows
  else:
    placement rows <- selected rows
    if t == 2 and sample/full representativeness fails:
      placement rows <- all rows

  evaluate model scores on analysis/placement rows

  calculate:
    preliminary exchangeability rank target
    10-standard-error heuristic target
    score-dependence design effect
    effective sample sizes
    final target mass tau*

  choose target order-statistic cutoff c*
  update run-location and probe histories
  calculate history reserve
  calculate structural/probe reserves
  extra_reserve <- maximum of all reserve terms

  verify scalar output bias if not already verified

  delta <- max(0, -c* + numerical_guard + extra_reserve)
  apply delta through verified scalar bias only
  cumulative_score_shift <- cumulative_score_shift + delta

  determine whether cross-run reserve is established
  ratchet required_depth if not established
  verify shifted target-rank cutoff

  if all release conditions hold:
    mark this observation GENERALIZATION_SAFE

```

5. Experimental Methodology

5.1 Evidence tiers and chronology

The evidence is divided into four tiers.

Tier A — matched controlled validation. The 13 original generator families belong to the general random-data environment used while the Tesla method was being developed. They are not treated as untouched external validation. Their purpose is controlled baseline comparison under matched randomness.

Tier B — independent-seed replication. The same 13 families are repeated at larger sample count using method-specific seeds. This tests numerical reproducibility but does not create a new distribution family.

Tier C — post-freeze challenge. Eight generator families absent from the development laboratory were

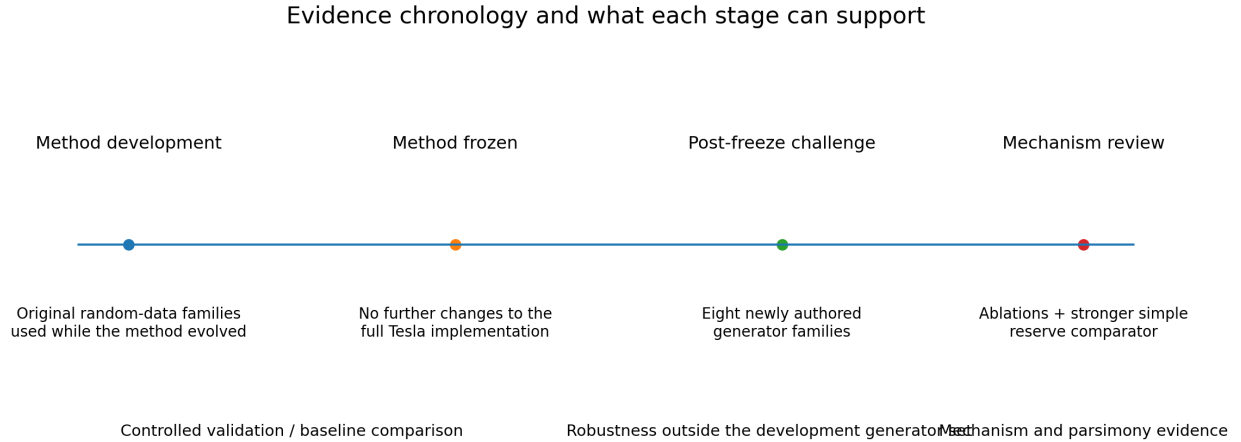


Figure 5. Evidence chronology. The original random-data families belonged to the broader development environment. The Tesla implementation was then frozen before the eight post-freeze challenge generators were authored. Ablations and the stronger simple reserve comparator were introduced later as mechanism and parsimony analyses.

created only after the Tesla implementation had been frozen. The method was not modified after observing these results. This was not preregistered. We therefore call it the **post-freeze challenge suite**: the algorithm was fixed before these distributions were designed, but the experiment was not externally preregistered.

Tier D — targeted failure and mechanism experiments. These include adversarial shift, component ablation, and the stronger conservative-reserve baseline. They are exploratory mechanism studies, with their chronology disclosed explicitly.

The chronology of development, freezing, post-freeze challenge construction, and later mechanism review is summarized in Figure~5. This separation is important because it determines which experiments can support confirmation claims and which are exploratory.

5.2 Software-independent numerical conventions

A faithful reimplementation should use:

- **PCG64 (Permuted Congruential Generator with 64-bit output)** for all NumPy-style experimental random streams. PCG64 is the pseudorandom-number generator used to make the seeded synthetic data reproducible;
- IEEE-754 `float32` feature matrices for model evaluation;
- `float64` accumulation for summary statistics where formulas specify it;
- a separate scalar pseudorandom-number-generator seed for every stream specified below;
- CPU evaluation for the reported experiments.

Exact package versions used for the reported run were:

```
Python 3.13.5
NumPy 2.3.5
PyTorch 2.10.0+cpu
SciPy 1.17.0
Linux 6.18.35 x86_64
AMD EPYC 9V74
CUDA: not used
```

An implementation in another language should reproduce the distributions and algorithms rather than rely on these libraries, but exact bitwise agreement of pseudorandom draws is only expected when the same pseudorandom-number generator and library versions are used.

5.3 Models

Linear model

For d features,

$$f(x) = w^\top x + b.$$

After setting the model seed, overwrite

$$w_j \sim \mathcal{N}(0, 0.02^2),$$

and set

$$b = -0.01.$$

Multilayer perceptron (MLP)

Use

$$x \rightarrow \text{Linear}(d, 32) \rightarrow \text{ReLU} \rightarrow \text{Linear}(32, 1).$$

With the model seed fixed before construction, initialize each linear layer as the PyTorch 2.10 `Linear` default, equivalent to independent uniform weights

$$W_{ij} \sim \mathcal{U}\left(-\frac{1}{\sqrt{\text{fan-in}}}, +\frac{1}{\sqrt{\text{fan-in}}}\right),$$

with the corresponding default uniform bias. After construction overwrite only the final scalar bias with

$$b_{\text{final}} = -0.01.$$

No ordinary model training occurs during the Tesla runtime.

5.4 Main run size and dimensions

Unless a sensitivity experiment says otherwise:

$$n = 5000, \quad d = 16.$$

Each trial uses 10 future holdout batches.

The release harness permits at most 20 observed batches and requires two release-qualified observations before terminating the observation loop. In the primary paired study Tesla reached its second release-qualified observation at observed batch 3 in every trial.

5.5 Original 13 run generators

All random choices below are redrawn independently for each run from that trial’s run generator.

iid — independent and identically distributed (IID) Gaussian

Each feature value is drawn independently from the same standard normal distribution:

$$X_{ij} \sim \mathcal{N}(0, 1).$$

heavy

$$X_{ij} = T_{ij}/\sqrt{3}, \quad T_{ij} \sim t_3.$$

block

Use block size 32. For each block and feature,

$$S_{bj} \sim \mathcal{N}(0, 0.90^2).$$

For each row,

$$\epsilon_{ij} \sim \mathcal{N}(0, 0.45^2),$$

$$X_{ij} = S_{\lfloor i/32 \rfloor, j} + \epsilon_{ij}.$$

mean_drift

Draw one run mean per feature,

$$\mu_j \sim \mathcal{N}(0, 0.45^2),$$

then

$$X_{ij} \sim \mathcal{N}(\mu_j, 1).$$

global_scale

Draw

$$s = \exp(G), \quad G \sim \mathcal{N}(0, 0.38^2),$$

then

$$X_{ij} = sZ_{ij}, \quad Z_{ij} \sim \mathcal{N}(0, 1).$$

anisotropic

For feature index $j = 0, \dots, d-1$, draw

$$g_1 \sim \mathcal{N}(0, 0.42^2), \quad g_2 \sim \mathcal{N}(0, 0.28^2),$$

and independent

$$e_j \sim \mathcal{N}(0, 0.12^2).$$

Define

$$a_j = g_1 \sin(0.7(j+1)) + g_2 \cos(1.3(j+1)) + e_j,$$

$$s_j = \exp(\text{clip}(a_j, -1.2, 1.2)).$$

Then

$$X_{ij} = s_j Z_{ij}, \quad Z_{ij} \sim \mathcal{N}(0, 1).$$

mixture

Draw

$$\mu_j \sim \mathcal{N}(0, 0.25^2), \quad p \sim U(0.30, 0.70).$$

For each row draw $C_i \sim \text{Bernoulli}(p)$. Generate

$$A_{ij} \sim \mathcal{N}(\mu_j - 0.50, 0.75^2),$$

$$B_{ij} \sim \mathcal{N}(\mu_j + 0.50, 0.75^2),$$

and select A_i when $C_i = 1$, otherwise B_i .

combo

Draw

$$\mu_j \sim \mathcal{N}(0, 0.35^2), \quad g \sim \mathcal{N}(0, 0.30^2), \quad e_j \sim \mathcal{N}(0, 0.10^2).$$

Define

$$s_j = \exp(\text{clip}(g \sin(0.9(j+1)) + e_j, -0.9, 0.9)).$$

For 32-row blocks draw

$$S_{bj} \sim \mathcal{N}(0, (0.55s_j)^2),$$

and row noise

$$E_{ij} \sim \mathcal{N}(\mu_j, (0.75s_j)^2).$$

Then

$$X_{ij} = S_{[i/32],j} + E_{ij}.$$

long_block

Draw one entire-run displacement

$$S_j \sim \mathcal{N}(0, 1.50^2),$$

and row noise

$$\epsilon_{ij} \sim \mathcal{N}(0, 0.03^2).$$

Then

$$X_{ij} = S_j + \epsilon_{ij}.$$

run_shift

Draw $v \sim \mathcal{N}(0, I_d)$ and normalize

$$u = v / \|v\|_2.$$

Draw scalar shift

$$a \sim \mathcal{N}(0, 1.50^2),$$

and noise

$$\epsilon_{ij} \sim \mathcal{N}(0, 0.20^2).$$

Then

$$X_i = au + \epsilon_i.$$

heavy_run_shift

Draw random unit direction u as above. Draw

$$a = 1.25T, \quad T \sim t_2,$$

and

$$\epsilon_{ij} \sim \mathcal{N}(0, 0.15^2).$$

Set

$$X_i = au + \epsilon_i.$$

rare_jump

Draw base run mean

$$\mu_j \sim \mathcal{N}(0, 0.25^2).$$

With probability 0.05 add

$$J_j \sim \mathcal{N}(0, 4^2)$$

to every feature mean.

Then

$$X_{ij} \sim \mathcal{N}(\mu_j, 0.85^2).$$

periodic_alias

Use period $P = 32$. Draw phase

$$\phi \sim \text{Unif}\{0, \dots, 31\},$$

amplitude

$$a \sim U(0.75, 2.0),$$

and a random unit direction u .

For row i ,

$$p_i = \sin\left(2\pi \frac{i + \phi}{32}\right).$$

Draw

$$\epsilon_{ij} \sim \mathcal{N}(0, 0.35^2),$$

and set

$$X_i = ap_i u + \epsilon_i.$$

5.6 Minimal bias baselines

One-run zero-mass

For one observed batch, compute scores z_i and apply

$$\Delta = \max(0, -\min_i z_i + 10^{-6}).$$

Freeze.

Three-run zero-mass

Repeat the same minimal operation sequentially on the first three observed batches, then freeze.

These controls do not use Tesla’s rank target, dependence correction, historical reserve, structural probes, or gate during state placement.

5.7 Primary matched experiment

Profile order is exactly:

```
0 iid
1 heavy
2 block
3 mean_drift
4 global_scale
5 anisotropic
6 mixture
7 combo
8 long_block
9 run_shift
10 heavy_run_shift
11 rare_jump
12 periodic_alias
```

For profile index p and within-profile replicate i , use model seed

$$s = 42000 + 100003p + 1009i + A,$$

where

$$A = \begin{cases} 0, & \text{linear,} \\ 700000003, & \text{MLP.} \end{cases}$$

Use:

- $i = 0, \dots, 5$ for linear models;
- $i = 0, \dots, 3$ for MLP models.

This produces

$$13(6 + 4) = 130$$

matched trial IDs.

For each trial, all methods receive the **same initial model seed**.

Generate up to 20 observed batches using PCG64 with seed

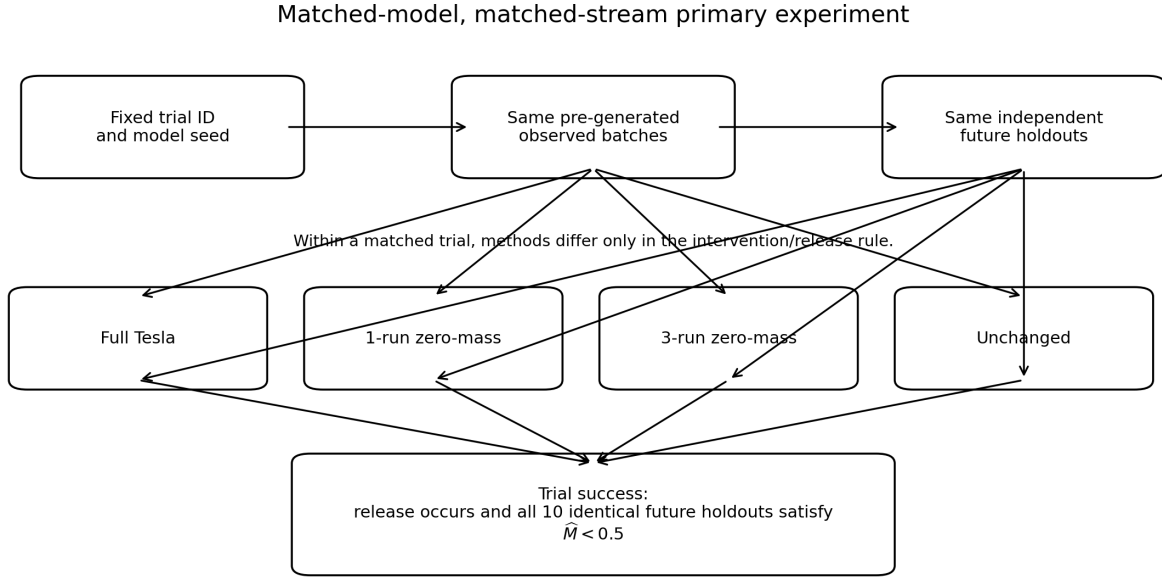


Figure 6. Matched-model, matched-stream primary experiment. Within a trial, all methods start from the same initialized model and receive the same observed and future batches. A trial succeeds only when release occurs and all ten identical future holdouts remain below the 0.5 canonical adverse-mass threshold.

$$s_{\text{train}} = s + 17.$$

Generate the 10 holdout batches from an independent PCG64 stream

$$s_{\text{holdout}} = s + 7,777,777.$$

This separation prevents Tesla’s variable release depth from changing which holdout batches are evaluated. The primary success event is

$$S = \mathbf{1}[\text{released} \wedge \max_{j=1,\dots,10} \widehat{M}_j < 0.5].$$

The endpoint is **unconditional future persistence after release**, not conditional-on-gate performance. In the paired experiment all 130 Tesla trials also had 100% gate acceptance on the ten ordinary same-process holdouts.

Figure~6 illustrates the paired experimental design used for the primary comparison.

5.8 Independent-seed replication

For architecture a , profile index p , method index m , and replicate i , use

$$s = 10000 + 1009i + 100003p + 10000019m + A,$$

with architecture offset A defined above.

Method order is:

```

0 tesla
1 one_run_zero
2 three_run_zero
3 unchanged

```

Use 12 seeds per profile for the linear model and 8 for the MLP.

Unlike the primary experiment, method seeds differ; this is why the independent replication is not used for paired inference.

5.9 External-label audits

For each independent-replication trial, initialize an audit random-number generator with

$$s_{\text{audit}} = s + 991337.$$

Draw a random vector

$$w_* \sim \mathcal{N}(0, I_d)$$

and normalize it.

For each holdout define unrelated linear labels

$$Y_i^{\text{lin}} = \mathbf{1}[X_i^\top w_* > 0],$$

and independent labels

$$Y_i^{\text{ind}} \sim \text{Bernoulli}(0.5).$$

The model audit prediction is

$$\hat{Y}_i = \mathbf{1}[f_\theta(X_i) > 0].$$

These labels are generated only after the Tesla state has been established and are never supplied to state placement or release.

5.10 Post-freeze challenge generators

All post-freeze challenge-suite runs use $n = 5000$, $d = 16$, and 10 holdouts.

skew_gamma

Draw

$$k \sim U(1.5, 4.0), \quad \theta = 1/\sqrt{k}.$$

Draw $\text{Gamma}(k, \theta)$, subtract its mean $k\theta$, and retain the resulting unit-variance positively skewed features.

lognormal_centered

Draw

$$\sigma \sim U(0.45, 0.9),$$

then $Z \sim \text{LogNormal}(0, \sigma^2)$. Standardize analytically using

$$E[Z] = e^{\sigma^2/2},$$

$$\text{Var}(Z) = (e^{\sigma^2} - 1)e^{\sigma^2}.$$

ar1_temporal — first-order autoregressive temporal dependence, **AR(1)**

Draw

$$\rho \sim U(0.85, 0.985).$$

Initialize $X_0 \sim N(0, I)$, then

$$X_t = \rho X_{t-1} + \sqrt{1 - \rho^2} \epsilon_t, \quad \epsilon_t \sim N(0, I).$$

Add one run-wide mean vector $N(0, 0.25^2 I)$.

rotating_covariance

Draw Gaussian matrix A , take an orthogonal–triangular (QR) matrix factorization $A = QR$, and draw eigenvalues

$$\lambda_j = e^{G_j}, \quad G_j \sim N(0, 0.65^2).$$

Normalize eigenvalues by their arithmetic mean. For $Z \sim N(0, I)$,

$$X = Z(Q \operatorname{diag}(\sqrt{\lambda}))^\top.$$

banana

Start with $Z \sim N(0, I)$. For $d \geq 2$, draw

$$c \sim U(0.35, 0.85)$$

and transform

$$Z_{i2} \leftarrow Z_{i2} + c(Z_{i1}^2 - 1).$$

Add a run-wide mean vector $N(0, 0.15^2 I)$.

sparse_run_shock

Start with row noise $N(0, 0.65^2 I)$. Draw $k \in \{1, 2, 3\}$ subject to $k \leq d$, select k features without replacement, and add independent $N(0, 2.5^2)$ run-wide shifts to those features.

two_scale_mixture

Draw

$$p \sim U(0.2, 0.8).$$

Each row is independently drawn from either

$$N(0, 0.35^2 I)$$

with probability p , or

$$N(0, 1.8^2 I)$$

otherwise.

squarewave_alias

Draw period

$$P \in \{17, 29, 47, 61\},$$

phase uniformly over $0, \dots, P-1$, amplitude

$$a \sim U(0.8, 2.2),$$

and random unit direction u .

Define

$$q_i = \begin{cases} +1, & (i + \phi) \bmod P < P/2, \\ -1, & \text{otherwise.} \end{cases}$$

Then

$$X_i = a q_i u + \epsilon_i, \quad \epsilon_i \sim N(0, 0.3^2 I).$$

For challenge profile index p , use

$$s = 6,100,000 + 100003p + 1009i + A,$$

with 10 linear and 6 MLP seeds per profile.

5.11 Cross-profile shift matrix

Training profiles:

```
iid
combo
mean_drift
```

Test profiles:

```
iid
combo
mean_drift
global_scale
long_block
run_shift
heavy_run_shift
rare_jump
```

Use 16 linear-model seeds per train/test pair:

$$s = 500000 + 1,000,003 p_{\text{train}} + 100003 p_{\text{test}} + 1009i.$$

Holdout-stream seed:

$$s_{\text{holdout}} = s + 7,000,001.$$

5.12 Adversarial failure-boundary experiment

Use 32 independent linear models with seeds

$$s_i = 900000 + 1009i, \quad i = 0, \dots, 31.$$

Establish the full Tesla method on independent-and-identically-distributed (IID) Gaussian runs with $n = 5000$, $d = 16$.

After release, let w be the frozen linear weight vector and define

$$u = w / \|w\|_2.$$

Use the evaluation random-number generator

$$s_{\text{adv}} = s_i + 333333.$$

At each tested magnitude

$$\delta \in \{0, 0.5, 1, 2, 4, 8, 12, 16, 24, 32, 48, 64, 96, 128, 192, 256\},$$

draw a fresh IID Gaussian batch Z_δ and construct

$$X_\delta = Z_\delta - \delta u.$$

For a linear score,

$$w^\top X_\delta + b = w^\top Z_\delta + b - \delta \|w\|_2,$$

so increasing δ deterministically pushes scores in the harmful direction.

Record first gate rejection and first canonical failure.

5.13 Sensitivity study

Test:

$$n \in \{500, 2000, 10000\},$$

$$d \in \{8, 32\},$$

on:

```
long_block
run_shift
heavy_run_shift
rare_jump
```

with six linear-model seeds per condition and six holdouts.

Seed formula:

$$s = 2,100,000 + 10,000,019 i_n + 1,000,003 i_d + 100003 i_p + 10007 i_m + 1009 i.$$

5.14 Exploratory component ablation

Use the same seeds and batches as the primary paired study, restricted to the four harsh profiles:

```
long_block
run_shift
heavy_run_shift
rare_jump
```

This creates 40 matched trials: 24 linear and 16 MLP.

Compare:

Full Tesla method.

Rank-only. For each of the first three batches, calculate only the exchangeability rank target and apply the corresponding direct scalar placement with zero extra reserve.

Noise-adjusted rank. Add Tesla’s dependence design effect and effective-sample-size target recalculation, but use zero history and structural reserve.

Three-run zero-mass. The minimal baseline from Section 5.6.

This ablation was added after the primary experiment and is treated as exploratory.

5.15 Strong conservative-reserve baseline

A reviewer-style stress check asked whether a large simple reserve could match the complete probe ensemble.

Several exploratory scale-multiplier baselines were tested on the 40 harsh matched trials. The strongest simple construction was:

1. for each of the first three batches, record the **pre-shift median score** m_t ;
2. minimally shift that batch to zero observed canonical mass;
3. after the third batch add

$$R_{\text{MR}} = 10 \left(\max_t m_t - \min_t m_t \right)$$

to the scalar score bias.

For brevity in equations only, the reserve is denoted R_{MR} . In prose we call the method the **ten-times median-range reserve comparator**.

The multiplier 10 was selected after exploratory comparison and is therefore **not confirmatory**.

After choosing the ten-times median-range reserve comparator, a new-seed confirmation used the same four harsh generator families with:

- 20 linear seeds per profile;

- 12 MLP seeds per profile;
- for 128 matched pairs of the full Tesla method and the ten-times median-range reserve comparator.
Seeds were:

$$s = 7,300,000 + 100003p + 1009i + A.$$

5.16 Ten-times median-range reserve plus the input-only gate

To determine whether the gate’s safety behavior depends on Tesla’s complex reserve, the ten-times median-range reserve comparator was also combined with the same input-only gate.

During the three comparator observation runs, store the same full-input mean/log-scale histories used by the Tesla input-only gate.

Then perform the same adversarial shift sweep as Section 5.12 on 32 new models with

$$s_i = 910000 + 1009i.$$

This simplified comparator uses neither Tesla’s structural probes nor Tesla’s rank target.

5.17 Statistical analysis

The independent experimental unit for the primary test is **one model trial**, not one row and not one holdout.

For the primary matched comparison, create binary trial-success vectors for Tesla and each baseline. For a baseline, let:

- b = number of pairs where Tesla succeeds and baseline fails;
- c = number where Tesla fails and baseline succeeds.

Because the methods are evaluated on the same trial IDs, use **McNemar’s paired-proportion test** [16], here evaluated exactly through its conditional binomial form rather than by a large-sample chi-square approximation:

$$p = \min(1, 2 \Pr[\text{Binomial}(b + c, 0.5) \leq \min(b, c)]).$$

The two planned comparisons—one-run and three-run zero-mass—are corrected by **Holm’s step-down procedure** [17], which controls the family-wise probability of at least one false rejection across the two planned tests.

For paired pass-rate differences, resample complete trial pairs with replacement 20,000 times and take the 2.5th and 97.5th percentiles.

For means of external-label accuracies, bootstrap complete **trials**, preserving the ten holdouts within each trial.

All bootstrap calculations use PCG64 with seed

$$20,260,815.$$

Observed binomial proportions are accompanied by **95% Wilson score confidence intervals** [18], an interval estimator for a binomial success probability, using

$$z = 1.959963984540054.$$

The post-freeze challenge suite is descriptive because both main methods achieve 100%.

The adversarial sweep is a repeated-measure path within each model. Its primary inferential unit is therefore the model-level ordering

$$\delta_{\text{reject}} \leq \delta_{\text{fail}},$$

not the 512 individual model/magnitude evaluations.

Exploratory ablation and ten-times median-range-reserve analyses are clearly labeled as post-primary and are not used to revise the confirmatory primary p -value family.

6. Results: Primary and Replication Experiments

6.1 Primary matched comparison

Tesla succeeded in every matched trial:

Method	Successful trials	Pass rate
Full Tesla method	130/130	100.0%
One-run zero-mass	112/130	86.2%
Three-run zero-mass	116/130	89.2%
Unchanged	17/130	13.1%

For Tesla versus one-run zero-mass:

$$b = 18, \quad c = 0.$$

The paired pass-rate advantage was

$$+0.13846,$$

with 95% paired-bootstrap interval

$$[0.08462, 0.20000].$$

The exact two-sided paired McNemar test [16] gave

$$p = 7.6294 \times 10^{-6},$$

and after Holm correction for the two planned comparisons [17],

$$p_{\text{Holm}} = 1.5259 \times 10^{-5}.$$

For Tesla versus three-run zero-mass:

$$b = 14, \quad c = 0.$$

The paired advantage was

$$+0.10769,$$

with 95% bootstrap interval

$$[0.06154, 0.16154].$$

The exact paired McNemar test gave

$$p = 1.2207 \times 10^{-4},$$

which remains

$$p_{\text{Holm}} = 1.2207 \times 10^{-4}$$

after Holm correction for the two planned comparisons [17].

The 95% Wilson score confidence interval [18] for 130/130 Tesla successes is

$$[0.9713, 1.0000].$$

Single-file note. The table and statistics above contain the complete numerical information that was originally plotted for this comparison.

6.2 Independent-seed replication

Model	Method	Successful trials	Failed trials	Worst future canonical mass
Linear	Full Tesla method	156/156	0	0.0002
Linear	One-run zero-mass	133/156	23	1.0000
Linear	Three-run zero-mass	139/156	17	1.0000
Linear	Unchanged	0/156	156	1.0000
MLP	Full Tesla method	104/104	0	0.0000
MLP	One-run zero-mass	90/104	14	1.0000
MLP	Three-run zero-mass	91/104	13	1.0000
MLP	Unchanged	32/104	72	1.0000

Combined full Tesla method:

260/260

successful trials and

2600/2600

future holdouts below 0.5.

95% Wilson score confidence intervals [18]:

260/260 : [0.98544, 1.0000],

2600/2600 : [0.99852, 1.0000].

All 260 Tesla trials had gate acceptance fraction 1.0 on their same-process future holdouts.

Complete linear-model per-profile replication results (12 trials per method and profile):

Distribution family	Full Tesla	One-run zero-mass	Three-run zero-mass	Unchanged
iid	12/12	12/12	12/12	0/12
heavy	12/12	12/12	12/12	0/12
block	12/12	12/12	12/12	0/12
mean drift	12/12	12/12	12/12	0/12
global scale	12/12	12/12	12/12	0/12
anisotropic	12/12	12/12	12/12	0/12
mixture	12/12	12/12	12/12	0/12
combo	12/12	12/12	12/12	0/12
long block	12/12	2/12	3/12	0/12
run shift	12/12	8/12	11/12	0/12
heavy run shift	12/12	4/12	5/12	0/12
rare jump	12/12	11/12	12/12	0/12
periodic alias	12/12	12/12	12/12	0/12

6.3 Where minimal baselines failed

Across long_block, run_shift, heavy_run_shift, and rare_jump in the independent replication:

Architecture	Tesla	One-run	Three-run
Linear	48/48	25/48	31/48
MLP	32/32	18/32	19/32
Combined	80/80	43/80	50/80

These distributions can move the score distribution coherently between runs, which is exactly the setting in which matching only the observed batch is insufficient.

6.4 Post-freeze challenge

Across eight generator families absent from the Tesla development laboratory:

128/128

Tesla trials succeeded.

The three-run zero-mass baseline also achieved

128/128.

The maximum future canonical mass was:

0.0000

for the full Tesla method and

0.2282

for the three-run baseline.

This result demonstrates breadth but **does not establish comparative superiority**.

7. Results: Scope, Failure Boundary, and Mechanism

7.1 Hidden-label negative controls

Against independent balanced labels:

$$\bar{A}_{\text{ind}} = 0.499953,$$

with a trial-bootstrap 95% confidence interval (CI)

$$[0.499675, 0.500235].$$

Against an unrelated hidden linear target:

$$\bar{A}_{\text{lin}} = 0.501933,$$

with 95% CI

$$[0.492983, 0.510864].$$

These results are compatible with chance and falsify the interpretation that controlling canonical mass creates generic label-free below-chance classification accuracy.

7.2 Cross-profile shift and conservative gate behavior

Across the train/test profile matrix, no ungated Tesla canonical failure occurred.

The input-only gate nevertheless rejected several large shifts. In particular, models established under ordinary processes often refused `long_block`, `run_shift`, and `heavy_run_shift` batches even though the scalar reserve still kept their canonical mass below 0.5.

The gate therefore behaves as a conservative **admissibility filter**, not as a calibrated probability-of-failure estimator.

7.3 Adversarial failure boundary for the full Tesla method

The model-aligned mean shift successfully breaks the frozen adverse state.

Across 32 independently released linear models:

Shift magnitude	Gate accepted	Ungated failures	Harmful accepted
0–16	32/32 at each level	0/32	0/32
24	29/32	1/32	0/32
32	0/32	32/32	0/32
48–256	0/32	32/32 at each level	0/32

First-rejection / first-failure ordering:

29 models: reject at 32, fail at 32
 2 models: reject at 24, fail at 32
 1 model : reject at 24, fail at 24

Thus

$$\delta_{\text{reject}} \leq \delta_{\text{fail}}$$

for

$$32/32$$

models.

The 95% Wilson score confidence interval [18] was:

$$[0.89282, 1.0000].$$

Across 285 accepted model/magnitude evaluations, none was harmful.

Single-file note. The adversarial result is fully specified by the table and first-rejection/first-failure counts above; no external figure is required.

The gate-versus-failure transition observed in the targeted adversarial shift experiment is summarized in Figure~7.

7.4 Sensitivity to sample size and dimension

Across the four harsh profiles, the full Tesla method passed all tested combinations of

$$n \in \{500, 2000, 10000\}$$

and

$$d \in \{8, 32\}.$$

The three-run minimal baseline remained unstable, especially under long-block dependence.

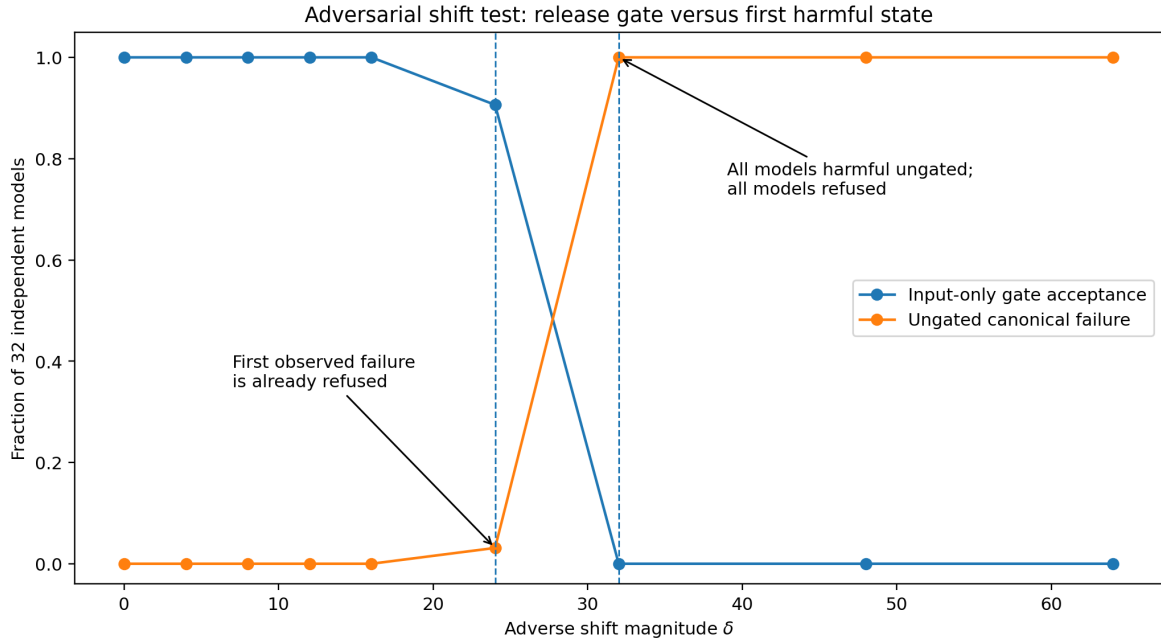


Figure 7. Adversarial shift experiment for 32 independently released linear models. The input-only gate begins refusing before or at the first observed harmful canonical state, and all models are refused by the shift magnitude at which every ungated model fails.

Complete sensitivity summary across the four harsh profiles (`long_block`, `run_shift`, `heavy_run_shift`, and `rare_jump`). Each cell aggregates four profiles $\times$ six trials = 24 trials:

Rows per batch	Features	Full Tesla successes	Three-run zero-mass successes
500	8	24/24	15/24
500	32	24/24	17/24
2,000	8	24/24	16/24
2,000	32	24/24	18/24
10,000	8	24/24	19/24
10,000	32	24/24	16/24

7.5 Component ablation

On 40 matched harsh trials:

Variant	Successful trials	Pass rate
Full Tesla method	40/40	100.0%
Rank-only	13/40	32.5%
Rank + dependence adjustment	14/40	35.0%
Three-run zero-mass	26/40	65.0%

Paired differences relative to the full Tesla method were:

$$+0.675$$

over rank-only, bootstrap 95% CI

$$[0.525, 0.825],$$

and

$$+0.650$$

over rank-plus-dependence, CI

$$[0.500, 0.800].$$

This demonstrates that the nominal rank target and dependence discount alone do not explain the full Tesla method’s observed robustness.

Single-file note. The ablation table above contains the complete values used in the corresponding plot; no external figure is required.

7.6 A stronger simple reserve nearly eliminates the gap

The initial exploratory sweep of simple oversized reserves showed that increasingly conservative scale multipliers improved success but still failed some harsh trials. The strongest simple rule used a reserve equal to ten times the range of the median model score across three observed runs; it passed 39/40 exploratory harsh trials.

Because the ten-times median-range rule was selected after that exploratory comparison, a second new-seed confirmation was run.

Across 128 matched pairs of the full Tesla method and the ten-times median-range comparator:

Architecture	Full Tesla method	Ten-times median-range comparator
Linear	80/80	80/80
MLP	48/48	48/48
Combined	128/128	128/128

Both methods had maximum future canonical mass 0.0 in this confirmation.

Therefore the current evidence **does not show that Tesla’s full structural-probe ensemble is necessary or superior to a sufficiently conservative simple empirical reserve.**

This negative/parsimony result materially changes the interpretation of the study.

Figure~8 summarizes the component-ablation result and the later parsimony finding.

7.7 The input-only gate remains effective with the simple reserve

The ten-times median-range comparator was then paired with the same input-only gate and evaluated on a new 32-model adversarial sweep.

For all 32 models,

$$\delta_{\text{reject}} \leq \delta_{\text{fail}},$$

and there were zero harmful accepted evaluations.

In this stronger-reserve system:

- the gate began refusing at magnitudes 24 or 32;
- failures began at magnitude 48 for most models and at 64 for the remaining models;
- gate acceptance was 0 by magnitude 32;
- failure rate was 90.625% at magnitude 48 and 100% by magnitude 64.

Thus the gate’s observed safety-precedence behavior is **separable from Tesla’s complex reserve construction.**

This is arguably the most important mechanism result in the paper: the tested safety architecture can be decomposed into a conservative one-sided reserve plus an input-only admissibility gate.

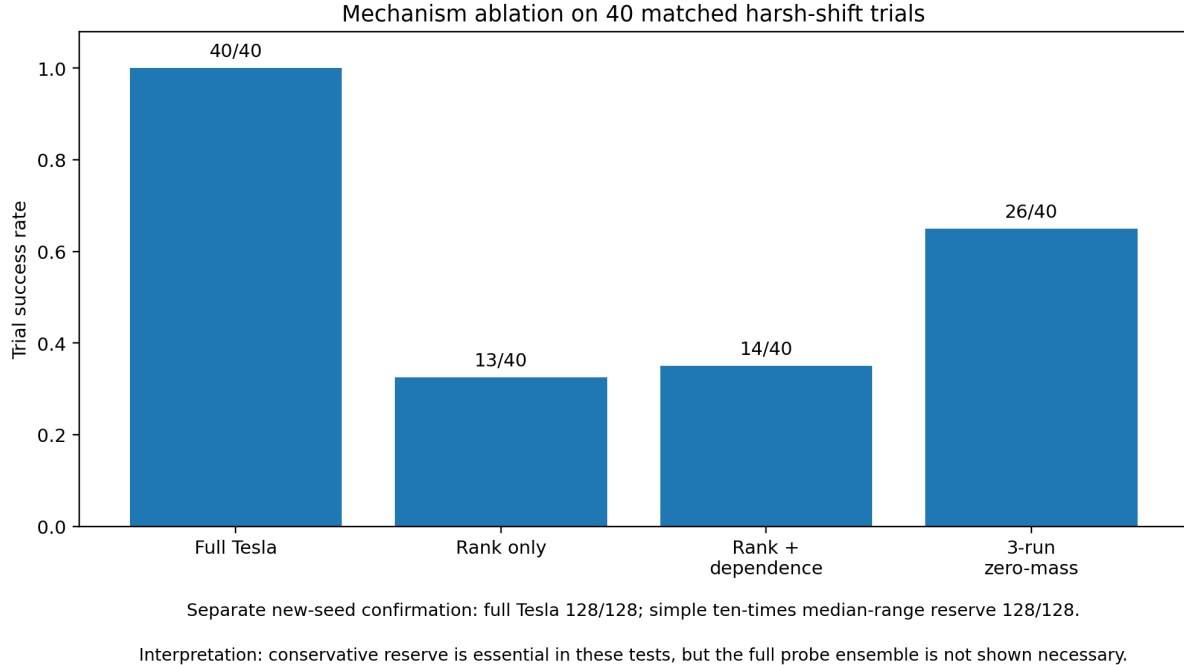


Figure 8. Mechanism and parsimony result. On 40 matched harsh-shift trials, rank placement and rank placement with dependence correction were insufficient, whereas the full method passed all trials. A later new-seed confirmation found that a much simpler ten-times median-range reserve also matched the full method on 128/128 trials, so the complete structural-probe ensemble is not shown to be necessary.

8. Discussion

8.1 What the experiments establish

The experiments establish three robust empirical facts within the tested synthetic setting.

First, minimal observed-run placement is not enough under coherent run-level displacement. The one-run and three-run zero-mass baselines fail repeatedly despite making their observed batches maximally adverse.

Second, conservative additional reserve fixes most or all of those failures. The full Tesla method succeeds, but so does a much simpler empirical between-run reserve in the expanded harsh confirmation.

Third, the input-only gate can reject a deliberately harmful covariate path before or at the first observed canonical failure. Importantly, this behavior persists when the complex reserve is replaced by the ten-times median-range reserve.

8.2 What the experiments do not establish

The paper does not show:

- generic hidden-label anti-learning;
- a universal 10^{-24} safety probability;
- a distribution-free robustness theorem;
- necessity of Tesla’s large structural-probe ensemble;
- superiority to every possible conservative scalar reserve;
- real-world application performance.

The hidden-label audit directly contradicts the first claim, and the strong-reserve baseline removes support for the fourth.

8.3 The likely essential architecture

The evidence suggests a simpler conceptual decomposition:

one-sided adverse placement + between-run reserve + input-only gate

rather than

all Tesla probe machinery is required.

The full Tesla implementation remains a robust realization of the first architecture, but its complexity should be treated as an engineering upper bound rather than a demonstrated minimal method.

8.4 Why the gate result matters

A large fixed bias can always be made more conservative, but no finite scalar reserve can guarantee safety against an unbounded adverse distribution shift.

This is visible in both adversarial experiments: sufficiently large shift eventually forces the frozen head back through the 0.5 boundary.

The gate addresses a qualitatively different problem. It says, in effect:

“The current input distribution has moved outside the region characterized by previous accepted runs; do not release.”

Because it receives only X , it does not need future labels or even a model forward pass.

8.5 Relation to testable deployment

The resulting architecture is closer to testable deployment and selective prediction than to classical anti-learning alone [7–11].

The anti-learning aspect determines the desired **direction** of the state. The selective-release literature provides the more relevant conceptual frame for deciding whether that state may be used under unlabeled shift.

8.6 Tesla-valve analogy revisited

The empirical distillation also makes the Tesla-valve metaphor cleaner.

The essential principle is not elaborate internal machinery for its own sake. It is **directional operation plus passive admissibility**.

The physical Tesla valve obtains direction-dependent resistance from geometry. The machine-learning system obtains a one-sided output state from scalar placement and uses input distribution geometry to decide whether operation should continue.

9. Limitations and Threats to Validity

9.1 Canonical mass is not arbitrary-label accuracy

This is the largest conceptual limitation and is demonstrated empirically, not merely acknowledged.

9.2 Main generators were known during development

The primary 13-family comparison is a controlled validation study. It should not be described as untouched external confirmation.

9.3 The post-freeze challenge suite was not preregistered

The challenge generators were created after the Tesla implementation was frozen, but the experiment was not externally preregistered. We therefore describe this evidence as a **post-freeze challenge**, not as a preregistered prospective test.

9.4 The ten-times median-range comparator is post hoc

The ten-times median-range comparator was selected after an exploratory strong-baseline sweep. Its later 128-pair confirmation uses new random seeds but the same harsh generator families. It therefore establishes reproducibility across seeds, not untouched distributional generalization.

A preregistered external comparison between the full Tesla method and a simple-reserve method is a priority for future work.

9.5 Synthetic evaluation

Synthetic processes are appropriate for controlled failure-path analysis but cannot substitute for a real task with independently justified canonical correctness semantics.

9.6 No universal release theorem

The input gate is empirically successful on the tested adversarial path. No theorem proves that every harmful distribution must cross the gate boundary first.

9.7 Conservative refusals

The gate sometimes refuses safe runs. Future work should report the full trade-off between:

$$P(\text{harmful accept})$$

and

$$P(\text{safe refuse}).$$

9.8 Small model classes

Only a linear model and a one-hidden-layer MLP are evaluated. Larger nonlinear models may expose failure geometries not represented here.

9.9 Scalar intervention

A scalar bias is auditable but cannot reshape the decision surface.

9.10 Finite experimental scale

Even 0/2,600 observed holdout failures does not imply zero true risk. Confidence intervals are reported precisely to avoid that interpretation.

10. Reproduction Protocol Without Source Code

A reader can reproduce the primary study from the paper alone.

10.1 Primary experiment

For each of the 13 profiles:

1. Generate six linear and four MLP trial seeds from Section 5.7.
2. Construct the model exactly as Section 5.3.
3. Pre-generate 20 observed batches using the PCG64 pseudorandom number generator with seed $s + 17$.
4. Independently pre-generate 10 future batches from seed $s + 7,777,777$.
5. Clone the identical initial model for:
 - full Tesla method;

- one-run zero-mass;
 - three-run zero-mass;
 - unchanged.
6. Apply each method exactly as Sections 4 and 5.6 specify.
 7. For Tesla, continue observed runs until two observations satisfy the complete release criterion (that is, they are **release-qualified**) or 20 observations have been processed.
 8. Freeze the resulting model.
 9. Evaluate all ten identical future batches for every method.
 10. A trial succeeds only if release occurred and all ten future canonical masses are strictly less than 0.5.
 11. Perform paired exact McNemar tests at the trial level.
 12. Use 20,000 paired bootstrap replicates with seed 20,260,815.
 13. Holm-correct the two planned Tesla-versus-minimal-baseline tests.

Expected primary counts:

Full Tesla method:	130 / 130
1-run zero-mass:	112 / 130
3-run zero-mass:	116 / 130
unchanged:	17 / 130

10.2 Expected paired statistics

Tesla vs 1-run:	
Tesla-only wins	= 18
baseline-only wins	= 0
exact two-sided McNemar p	= 7.62939453125e-06
Holm p	= 1.52587890625e-05
pass-rate difference	= 0.1384615385
95% paired-bootstrap CI	= [0.0846153846, 0.2000000000]
Tesla vs 3-run:	
Tesla-only wins	= 14
baseline-only wins	= 0
exact two-sided McNemar p	= 0.0001220703125
Holm p	= 0.0001220703125
pass-rate difference	= 0.1076923077
95% paired-bootstrap CI	= [0.0615384615, 0.1615384615]

10.3 Independent replication

Use the method-specific seed formula from Section 5.8.

Expected counts:

Linear Tesla:	156 / 156
MLP Tesla:	104 / 104
Combined:	260 / 260
Future holdouts:	2600 / 2600 below 0.5

10.4 Hidden-label audit

Expected mean accuracies:

Independent balanced labels:	
mean	= 0.4999530769
95% trial-bootstrap CI	= [0.4996749962, 0.5002349231]
Unrelated linear labels:	
mean	= 0.5019331538
95% trial-bootstrap CI	= [0.4929832942, 0.5108637192]

10.5 Adversarial test of the full Tesla method

Use Section 5.12 exactly.

Expected first-boundary counts:

```
29 models: first reject 32, first fail 32
2 models: first reject 24, first fail 32
1 model : first reject 24, first fail 24
```

Expected safety precedence:

```
32 / 32 models
```

10.6 Component ablation

Use the primary matched seeds restricted to the four harsh profiles.

Expected results:

```
Full Tesla method:      40 / 40
Rank-only:             13 / 40
Rank + dependence:     14 / 40
3-run zero-mass:       26 / 40
```

10.7 Strong baseline confirmation

Use Section 5.15.

Expected results:

```
Linear:
  Full Tesla method = 80 / 80
  Ten-times median-range comparator = 80 / 80

MLP:
  Full Tesla method = 48 / 48
  Ten-times median-range comparator = 48 / 48

Combined:
  Full Tesla method = 128 / 128
  Ten-times median-range comparator = 128 / 128
```

10.8 Ten-times median-range comparator plus gate: adversarial test

Use Section 5.16.

Expected:

```
Gate-before-failure precedence: 32 / 32
Harmful accepted evaluations: 0
Gate acceptance becomes 0 by magnitude 32.
Failure rate = 0 through magnitude 32.
Failure rate = 0.90625 at magnitude 48.
Failure rate = 1.0 by magnitude 64.
```

11. Full Numerical Parameter Specification

The following table lists every numerical setting used by the frozen Tesla implementation in the experiments. The left column retains the original implementation label only so an independent reimplementation can map each setting unambiguously to the equations above; **the reader is not expected to infer meaning from these labels**. Each group heading states the scientific role of the parameters. Settings from obsolete development logic that is not executed by the method reported here are omitted.

Bias verification / placement

Constant	Value
TESLA_ANTI_TRAIN_LOGIT_MARGIN	0.0
TESLA_DIRECT_LOGIT_GUARD	1e-06
TESLA_OUTPUT_BIAS_LINEARITY_MULTIPLIER	16.0
TESLA_OUTPUT_BIAS_LINEARITY_REL_TOL	0.002
TESLA_OUTPUT_BIAS_PROBE_EPS	0.001
TESLA_OUTPUT_BIAS_PROBE_ROWS	64
TESLA_OUTPUT_BIAS_UNIFORM_TOL	0.0001
TESLA_PREDICTED_POST_ROW_GUARD	2

Rank target / dependence

Constant	Value
TESLA_BELOW_THRESHOLD_CUSHION	0.005
TESLA_GENERALIZATION_MAX_CUSHION	0.22
TESLA_RUNTIME_FUTURE_Z	10.0
TESLA_RUNTIME_GENERALIZATION_Z	10.0
TESLA_RUNTIME_NOISE_AUTOCORR_LAGS	8
TESLA_RUNTIME_NOISE_AUTOCORR_SHRINK	0.02
TESLA_RUNTIME_NOISE_BASE_DESIGN_EFFECT	1.15
TESLA_RUNTIME_NOISE_BLOCK_MAX	256
TESLA_RUNTIME_NOISE_BLOCK_MIN	16
TESLA_RUNTIME_NOISE_BLOCK_SHRINK	0.75
TESLA_RUNTIME_NOISE_FLOOR_CUSHION	0.03
TESLA_RUNTIME_NOISE_MAX_DESIGN_EFFECT	8.0
TESLA_RUNTIME_TOTAL_CROSSRUN_RISK	1e-24

Sampling / release geometry

Constant	Value
TESLA_RELEASE_SAMPLE_LOG_SCALE_MAX_MAX	1.25
TESLA_RELEASE_SAMPLE_LOG_SCALE_RMS_MAX	0.35
TESLA_RELEASE_SAMPLE_MEAN_MAX_MAX	1.25
TESLA_RELEASE_SAMPLE_MEAN_RMS_MAX	0.35
TESLA_RUNTIME_BASE_PROCESSED_ROWS	320
TESLA_RUNTIME_FULL_OBSERVATION_MILESTONES	(3, 4, 5, 6, 8, 10, 12)
TESLA_RUNTIME_MAX_PROCESSED_ROWS	1024
TESLA_RUNTIME_SAMPLE_WINDOW_ROWS	16
TESLA_RUNTIME_SERIAL_BLOCKS	32
TESLA_RUNTIME_SERIAL_DESIGN_TRIGGER	2.5

Input-only gate

Constant	Value
TESLA_INPUT_RUN_GATE_BETWEEN_FLOOR	1.5
TESLA_INPUT_RUN_GATE_ENABLED	True

Constant	Value
TESLA_INPUT_RUN_GATE_HISTORY	8
TESLA_INPUT_RUN_GATE_LOG_SCALE_FLOOR	0.25
TESLA_INPUT_RUN_GATE_MEAN_MAX_Z	10.0
TESLA_INPUT_RUN_GATE_MEAN_RMS_Z	4.5
TESLA_INPUT_RUN_GATE_MIN_HISTORY	2
TESLA_INPUT_RUN_GATE_RADIAL_BETWEEN_MULTIPLIER	32.0
TESLA_INPUT_RUN_GATE_SCALE_MAX_Z	10.0
TESLA_INPUT_RUN_GATE_SCALE_RMS_Z	5.0
TESLA_INPUT_RUN_GATE_SUMMARY_ROWS	128

Clarification. The value 128 in the row above is **not** the number of rows used by the input-only acceptance gate. It is used only by a separate model-aware reserve calculation. The input-only gate defined in Sections 4.3–4.4 computes its feature mean and log-scale summaries from **every row in the current batch**.

Between-run history

Constant	Value
TESLA_RUNTIME_ADVERSE_TAIL_EXTRA_SIGMA	4.0
TESLA_RUNTIME_ADVERSE_TAIL_MIN_OBSERVATIONS	2
TESLA_RUNTIME_ADVERSE_TAIL_STABILITY_FRACTION	0.2
TESLA_RUNTIME_ADVERSE_TAIL_STABLE_STREAK	1
TESLA_RUNTIME_BETWEEN_RUN_SIGMA_Z	10.0
TESLA_RUNTIME_FEATURE_MEAN_HISTORY	8
TESLA_RUNTIME_HISTORY_MAX_OBSERVATIONS	16

Feature-drift probes

Constant	Value
TESLA_RUNTIME_FEATURE_DIAG_SCALE_DIRECTIONS	2
TESLA_RUNTIME_FEATURE_DIAG_SCALE_RADIUS_MAX	4.0
TESLA_RUNTIME_FEATURE_DIAG_SCALE_SE_TRIGGER	2.0
TESLA_RUNTIME_FEATURE_DIAG_SCALE_SIGMA_Z	8.0
TESLA_RUNTIME_FEATURE_DRIFT_DIRECTIONS	2
TESLA_RUNTIME_FEATURE_DRIFT_PROBE_ROWS	16
TESLA_RUNTIME_FEATURE_DRIFT_SIGMA_Z	8.0
TESLA_RUNTIME_FEATURE_MEAN_SAMPLE_ROWS	128
TESLA_RUNTIME_FEATURE_MEAN_SE_TRIGGER	1.5
TESLA_RUNTIME_FEATURE_SCALE_LOG_TRIGGER	0.05
TESLA_RUNTIME_FEATURE_SCALE_PRIOR_BASE	0.0
TESLA_RUNTIME_FEATURE_SCALE_PRIOR_EXTRA	0.5
TESLA_RUNTIME_FEATURE_SCALE_PRIOR_REFERENCE_ROWS	256
TESLA_RUNTIME_FEATURE_SCALE_RADIUS_MAX	6.0
TESLA_RUNTIME_FEATURE_SCALE_SIGMA_Z	10.0

Covariance probes

Constant	Value
TESLA_RUNTIME_COV_LOG_TRIGGER	0.1
TESLA_RUNTIME_COV_PCA_DIRECTIONS	2
TESLA_RUNTIME_COV_PCA_EIGEN_TRIGGER	2.75
TESLA_RUNTIME_COV_PCA_RADIUS_MAX	2.5
TESLA_RUNTIME_COV_PROBE_DIRECTIONS	2
TESLA_RUNTIME_COV_PROBE_ROWS	16
TESLA_RUNTIME_COV_PROJECTION_COUNT	6
TESLA_RUNTIME_COV_RADIUS_MAX	3.0
TESLA_RUNTIME_COV_SIGMA_Z	8.0

Gate-envelope probe

Constant	Value
TESLA_RUNTIME_GATE_ENVELOPE_PROBE_ROWS	12
TESLA_RUNTIME_GATE_ENVELOPE_RADIUS_FRACTION	1.0

Support probes

Constant	Value
TESLA_RUNTIME_HASH_SUPPORT_PROBE_ROWS	4096
TESLA_RUNTIME_SUPPORT_ELITE_RADIUS	0.8
TESLA_RUNTIME_SUPPORT_ELITE_ROWS	16
TESLA_RUNTIME_SUPPORT_ELITE_STEP	0.18
TESLA_RUNTIME_SUPPORT_ELITE_STEPS	4
TESLA_RUNTIME_SUPPORT_PROBE_ROWS	4096

Marginal recombination

Constant	Value
TESLA_RUNTIME_MARGINAL_ELITE_RADIUS	4.0
TESLA_RUNTIME_MARGINAL_ELITE_ROWS	64
TESLA_RUNTIME_MARGINAL_ELITE_STEP	0.65
TESLA_RUNTIME_MARGINAL_ELITE_STEPS	5
TESLA_RUNTIME_MARGINAL_RECOMBINE_ROWS	4096

Shape-stress probe

Constant	Value
TESLA_RUNTIME_SHAPE_STRESS_MAX_R2	2.2
TESLA_RUNTIME_SHAPE_STRESS_ROWS	768
TESLA_RUNTIME_SHAPE_STRESS_SPARSE_FRACTION	0.25
TESLA_RUNTIME_SHAPE_STRESS_SPARSE_SHARE	0.25

Model-direction probes

Constant	Value
TESLA_RUNTIME_MODEL_DIR_MAX	6
TESLA_RUNTIME_MODEL_DIR_POINTS	160
TESLA_RUNTIME_MODEL_DIR_RADIUS	3.5
TESLA_RUNTIME_MODEL_DIR_REFINE_POINTS	48
TESLA_RUNTIME_MODEL_PAIR_DIRS	4
TESLA_RUNTIME_MODEL_PAIR_GRID	25
TESLA_RUNTIME_MODEL_PAIR_RADIUS	3.5
TESLA_RUNTIME_MODEL_PAIR_REFINE	9

Opaque-corner probe

Constant	Value
TESLA_RUNTIME_OPAQUE_CORNER_BLOCK	16
TESLA_RUNTIME_OPAQUE_CORNER_MAX_FEATURES	20
TESLA_RUNTIME_OPAQUE_CORNER_RADIUS	0.8
TESLA_RUNTIME_OPAQUE_CORNER_SKIP_MARGIN	8.0

Between-run spherical probe

Constant	Value
TESLA_RUNTIME_BETWEEN_SPHERE_MIN_FRACTION	0.1
TESLA_RUNTIME_BETWEEN_SPHERE_ROWS	768
TESLA_RUNTIME_BETWEEN_SPHERE_WITHIN_RATIO_TRIGGER	1.0

Affine interaction screen

Constant	Value
TESLA_RUNTIME_AFFINE_INTERSECTION_RADIUS	6.0
TESLA_RUNTIME_AFFINE_SCREEN_HIGH_SLACK	2
TESLA_RUNTIME_AFFINE_SCREEN_HIGH_UNITS	12
TESLA_RUNTIME_AFFINE_SCREEN_MAX_ORDER	20
TESLA_RUNTIME_AFFINE_SCREEN_PAIR_EPS	0.1
TESLA_RUNTIME_AFFINE_SCREEN_QUAD_UNITS	10
TESLA_RUNTIME_AFFINE_SCREEN_QUINT_UNITS	8
TESLA_RUNTIME_AFFINE_SCREEN_REFINE_CANDIDATES	48
TESLA_RUNTIME_AFFINE_SCREEN_TRIPLE_UNITS	14
TESLA_RUNTIME_AFFINE_SCREEN_UNITS	24

Other runtime

Constant	Value
TESLA_RUNTIME_LOGIT_MARGIN_EXTRA	1.1
TESLA_RUNTIME_LOGIT_MARGIN_FACTOR	0.0
TESLA_RUNTIME_LOGIT_MARGIN_REFERENCE_ROWS	256
TESLA_RUNTIME_MAX_ADAPTIVE_OBSERVATIONS	12
TESLA_RUNTIME_MIN_NOISE_OBSERVATIONS	2

12. Data and Materials Availability

No human, clinical, proprietary, or personally identifiable dataset is required.

Every experimental input distribution is fully specified mathematically in Sections 5.5 and 5.10. Every trial seed and stream transformation is specified in Sections 5.7–5.16. Therefore the raw feature matrices can be regenerated independently.

The manuscript intentionally does not require access to the original Python files for methodological reproduction. Source-code hashes are retained only as provenance for the frozen implementation that generated the reported results:

```
Frozen Tesla implementation (the development repository happened to label this snapshot **revision 281**; that
    number has no theoretical or algorithmic meaning and is included only to identify the exact frozen snapshot)
:
8edd7bb70644e55774dff1c36ad5912fb72929cc0994fe7621a9d6e967191dd3

Standalone Tesla random-data development laboratory:
dcf77b38c8e2f701d42c7c12d1102e56964f2ee0aa7a1e256154c622a6ed0f14
```

Independent reimplementations should be judged against the expected aggregate values in Section 10 rather than against access to these files.

13. Conclusion

This paper began with the idea of **controlled anti-learning** but the experiments sharpened that idea substantially.

The full Tesla method reliably established and preserved a canonical adverse state across the tested synthetic runs. It significantly outperformed minimal one- and three-observation bias placement in a matched primary experiment and reproduced that behavior across a larger independent-seed campaign.

However, stronger mechanism tests show that the most complicated interpretation is not justified. A simple empirical between-run reserve can match the full Tesla method on the tested harsh processes, and the same input-only gate retains gate-before-failure precedence when attached to that simple reserve.

The evidence therefore supports a more parsimonious conclusion:

The experimentally supported Tesla principle is conservative one-sided adverse-state placement plus an input-only selective-release gate—not the necessity of a particular large probe ensemble.

This is still distinct from generic hidden-label anti-learning. External labels remain at chance, so a future application must independently establish the semantic orientation that turns canonical adverse mass into actual below-chance accuracy.

The next decisive experiment is therefore not another synthetic optimization of this frozen Tesla implementation. It is a preregistered, externally reproducible task in which:

1. correctness orientation is defined independently of deployment labels;
2. true hidden-label accuracy can be scored only after predictions are committed;
3. the release gate remains label-free;
4. the full Tesla method and a parsimonious reserve-plus-gate comparator are both frozen in advance;
5. harmful acceptance and safe refusal are both measured.

That experiment would determine whether the Tesla principle is merely a controlled synthetic phenomenon or a general method for selective deployment of canonical anti-learning states.

Declarations

Human or animal subjects: None.

Personal data: None.

Proprietary data: None.

Funding: To be supplied by the authors if applicable.

Competing interests: To be supplied before submission.

Author contributions: To be supplied after the author list is finalized.

Use of automated tools: Computational experiment generation and manuscript preparation used automated software assistance. The authors are responsible for verification of all reported methods, results, and claims.

References

Reference notation. DOI means **Digital Object Identifier**, a persistent identifier for a published work.

- [1] Kowalczyk, A., & Chapelle, O. (2005). An Analysis of the Anti-learning Phenomenon for the Class Symmetric Polyhedron. In *Proceedings of the 16th International Conference on Algorithmic Learning Theory (ALT 2005)*, Lecture Notes in Artificial Intelligence, vol. 3734, pp. 78–91. Springer. DOI: 10.1007/11564089_8.
- [2] Lisitsa, A., & Vernitski, A. (2024). On Strong Anti-learning of Parity. In *Optimization, Learning Algorithms and Applications: Third International Conference, OL2A 2023, Revised Selected Papers, Part II*, Communications in Computer and Information Science, vol. 1982, pp. 271–279. Springer. DOI: 10.1007/978-3-031-53036-4_19.
- [3] Cacioli, J.-P. (2026). Below-Chance Blindness: Prompted Underperformance in Small LLMs Produces Positional Bias Rather than Answer Avoidance. *arXiv:2604.25249*. Preprint.
- [4] Donmez, P., Lebanon, G., & Balasubramanian, K. (2010). Unsupervised Supervised Learning I: Estimating Classification and Regression Errors without Labels. *Journal of Machine Learning Research*, 11, 1323–1351.
- [5] Jaffe, A., Nadler, B., & Kluger, Y. (2015). Estimating the Accuracies of Multiple Classifiers without Labeled Data. *Proceedings of the Eighteenth International Conference on Artificial Intelligence and Statistics, Proceedings of Machine Learning Research*, vol. 38, pp. 407–415.
- [6] Balasubramanian, K., Donmez, P., & Lebanon, G. (2011). Unsupervised Supervised Learning II: Margin-Based Classification Without Labels. *Journal of Machine Learning Research*, 12, 3119–3145.
- [7] Kalai, A. T., & Kanade, V. (2021). Efficient Learning with Arbitrary Covariate Shift. *Proceedings of the 32nd International Conference on Algorithmic Learning Theory, Proceedings of Machine Learning Research*, vol. 132, pp. 850–864.
- [8] Klivans, A., Stavropoulos, K., & Vasilyan, A. (2024). Testable Learning with Distribution Shift. *Proceedings of the 37th Conference on Learning Theory, Proceedings of Machine Learning Research*, vol. 247, pp. 2887–2943.
- [9] Pouget, A., Yaghini, M., Rabanser, S., & Papernot, N. (2025). Suitability Filter: A Statistical Framework for Classifier Evaluation in Real-World Deployment Settings. *Proceedings of the 42nd International Conference on Machine Learning, Proceedings of Machine Learning Research*, vol. 267, pp. 49603–49627.
- [10] Timans, A., Verma, R., Nalisnick, E., & Naesseth, C. A. (2025). On Continuous Monitoring of Risk Violations under Unknown Shift. *Proceedings of the Forty-first Conference on Uncertainty in Artificial Intelligence, Proceedings of Machine Learning Research*, vol. 286, pp. 4204–4226.
- [11] Adil, D., & Blasiok, J. (2026). Efficient and Provable Algorithms for Covariate Shift. *Proceedings of the 37th International Conference on Algorithmic Learning Theory, Proceedings of Machine Learning Research*, vol. 313, pp. 1–34.
- [12] Tesla, N. (1920). *Valvular Conduit*. U.S. Patent 1,329,559. Filed February 21, 1916; patented February 3, 1920.
- [13] Nguyen, Q. M., Huang, D., Zauderer, E., Romanelli, G., Meyer, C. L., & Ristroph, L. (2021). Tesla’s Fluidic Diode and the Electronic-Hydraulic Analogy. *American Journal of Physics*, 89(4), 393–402. DOI: 10.1119/10.0003395.
- [14] Nguyen, Q. M., Abouezzi, J., & Ristroph, L. (2021). Early Turbulence and Pulsatile Flows Enhance Diodicity of Tesla’s Macrofluidic Valve. *Nature Communications*, 12, 2884. DOI: 10.1038/s41467-021-23009-y.
- [15] Serfling, R. J. (1974). Probability Inequalities for the Sum in Sampling without Replacement. *The Annals of Statistics*, 2(1), 39–48. DOI: 10.1214/aos/1176342611.
- [16] McNemar, Q. (1947). Note on the Sampling Error of the Difference Between Correlated Proportions or Percentages. *Psychometrika*, 12(2), 153–157. DOI: 10.1007/BF02295996.
- [17] Holm, S. (1979). A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.

[18] Wilson, E. B. (1927). Probable Inference, the Law of Succession, and Statistical Inference. *Journal of the American Statistical Association*, 22(158), 209–212. DOI: 10.1080/01621459.1927.10502953.

Supplementary Methods

S1. Deterministic 64-bit mixing function

Several row-selection and projection constructions use the same unsigned 64-bit mixer.

For input integer x , arithmetic is modulo 2^{64} . In the pseudocode below, **XOR** means bitwise exclusive OR and $\gg$ means an unsigned right shift:

```
x = x + 0x9E3779B97F4A7C15
x = (x XOR (x >> 30)) * 0xBF58476D1CE4E5B9
x = (x XOR (x >> 27)) * 0x94D049BB133111EB
x = x XOR (x >> 31)
```

All intermediate values are masked to 64 bits.

S1.1 Fixed stratified row index

To choose k deterministic positions among n rows:

1. set

$$\text{salt} = \text{mix64}(n \oplus 0xA0761D6478BD642F);$$

2. for stratum $i = 0, \dots, k - 1$:

$$lo = \lfloor in/k \rfloor, \quad hi = \lfloor (i+1)n/k \rfloor;$$

3. select

$$lo + \text{mix64}(\text{salt} + i) \bmod \max(1, hi - lo).$$

S1.2 Runtime contiguous-window row selection

For observation index o , stratum i , and current row cap:

$$h = \text{mix64}((o+1)0xD1B54A32D192ED03 + i).$$

Within that stratum choose a contiguous 16-row window starting at the allowed position indexed by h modulo the number of legal starts.

S1.3 Fixed sign projections

For projection j and feature q ,

$$P_{jq} = \frac{21[\text{mix64}((j+1)0xD1B54A32D192ED03 + q) \text{ is odd}] - 1}{\sqrt{d}}.$$

S2. Complete structural-probe specification

All probe margins have the same final form. If c_* is the target-rank cutoff and $z_{\min}^{\text{probe}}$ is the smallest score found by the probe,

$$R_{\text{probe}} = \max(0, c_* - z_{\min}^{\text{probe}} + 10^{-6}).$$

A probe can only increase the scalar reserve.

S2.1 Feature-mean and feature-scale drift probe

Use 128 deterministic stratified rows from each run.

For current sample mean μ_t , keep the last 8 run means. If at least two are present:

1. stack them;
2. subtract their mean;
3. compute a singular value decomposition (SVD);
4. retain at most the first two right-singular directions.

For direction u , project every historical mean. Let range be R_u . Let

$$R_{\text{noise}} = s_{\text{RMS}} \sqrt{2d/128},$$

where s_{RMS} is the current overall feature RMS.

Discard the direction unless

$$R_u > 1.5R_{\text{noise}}.$$

For retained directions estimate

$$\sigma_u = R_u/d_2(h),$$

and use displacement radius

$$r_u = |p_{\text{current}} - \bar{p}| + 8\sigma_u.$$

Probe both $\pm r_u u$ on the first 16 model-evaluation rows.

Also maintain per-feature log-RMS histories. SVD these histories and retain up to two anisotropic-scale directions. Approximate selected-direction noise by

$$\sqrt{d/(2 \cdot 128)}.$$

Require range larger than twice this value. Use radius

$$r_\ell = \min(4, |p_{\text{current}} - \bar{p}| + 8\sigma).$$

Probe multiplicative scales

$$\exp(\pm r_\ell u_j)$$

around the current feature mean.

Maintain global log-RMS history. A prior global scale radius is

$$r_{\text{prior}} = 0.5 \min(1, \sqrt{256/m}),$$

and historical global-scale radius, when history range exceeds 0.05, is

$$r_{\text{hist}} = \min(6, |\ell_{\text{current}} - \bar{\ell}| + 10\sigma_\ell).$$

Probe both expansion and contraction at the larger of prior and historical radii.

S2.2 Covariance projection probe

Standardize 128 deterministic rows feature-wise.

Use six fixed sign projections from S1.3. For each projection store

$$\log \sqrt{E[(ZP_j)^2]}.$$

With at least two runs of history, estimate range-based sigma and ignore projection histories whose log-range is at most 0.1.

For each projection,

$$r_j = \min(3, |\ell_j - \bar{\ell}_j| + 8\sigma_j).$$

Probe the two largest positive radii, in both expansion and contraction directions, using 16 deterministic rows.

Also perform a principal-component analysis (PCA) of the standardized 128-row sample by computing its covariance eigenvalues and eigenvectors. Keep the top two eigenvectors. If the largest historically observed leading eigenvalue exceeds 2.75, define

$$r_{\text{PCA}} = \min(2.5, 0.5 \log(\lambda_{\max}/2.75))$$

and probe expansion/contraction along the top two PCA directions.

S2.3 Gate-envelope score probe

With at least two input-gate histories, reconstruct the exact gate mean and log-scale standardizers.

Use 12 deterministic rows and compute the input gradient of the scalar score.

Map the gate's allowed RMS/box constraints into an adverse standardized displacement vector.

For a sensitivity vector s , RMS radius r , and per-coordinate box b :

1. desired Euclidean radius is $r\sqrt{d}$;
2. move opposite s ;
3. clip coordinates to $[-b, b]$;
4. redistribute remaining radius among unclipped coordinates;
5. repeat clipping/redistribution four times.

Use:

$$r_{\mu} = 4.5, \quad b_{\mu} = 10,$$

$$r_{\ell} = 5, \quad b_{\ell} = 10.$$

Evaluate:

- mean-only adverse probe;
- scale-only adverse probe;
- joint mean-plus-scale adverse probe.

S2.4 Actual-support probe

Create a fixed real-row support set from the union of:

- up to 4,096 deterministic stratified row positions;
- up to 4,096 rows with the smallest deterministic feature hashes.

Feature hashing reinterprets each float32 feature as a 32-bit unsigned integer. Initialize row hash

$$h = \text{0xCB F29CE484222325}.$$

For feature index j , update with a deterministic Fowler–Noll–Vo (FNV)-style hash mixing sequence using constants

```
0x9E3779B1
0x9E3779B97F4A7C15
0x100000001B3
```

exactly in unsigned 64-bit arithmetic.

Evaluate all fixed support rows. Select the 16 lowest-scoring support points **only after the real support set is fixed**.

Standardize each elite point relative to full-run mean and scale. Starting at its observed standardized location, take four score-gradient descent steps:

$$\eta_t = 0.18/(1 + 0.20t),$$

projecting displacement from the original elite point into an L_2 radius of 0.8.

The lowest score among original and refined support points defines the support reserve.

S2.5 Product-of-marginals recombination probe

Create 4,096 synthetic rows. For each feature column independently, sample values with replacement from that feature’s empirical values in the actual run.

Use deterministic PCG64 seed

$$[0xB7E15162 \oplus (d \cdot 0x9E3779B1) \oplus ((t + 1) \cdot 0x85EBCA77) \oplus n] \bmod 2^{32}.$$

Evaluate the 4,096 synthetic rows.

Take the 64 lowest-scoring rows. Standardize relative to full-run mean and scale, clip each standardized coordinate to $[-4, 4]$, and take five gradient-descent steps:

$$\eta_t = 0.65/(1 + 0.25t).$$

Normalize each gradient direction before stepping. If a gradient is numerically zero, replace it by a random unit direction from the same deterministic random-number generator.

Clip all standardized coordinates back to $[-4, 4]$ after every step.

S2.6 Nonlinear shape-stress probe

Generate 768 synthetic shell points plus the exact feature mean.

Use:

- 75% dense shell points;
- 25% sparse shell points;
- maximum standardized squared shell radius $r_{\max}^2 = 2.2$;
- sparse points modify 25% of features.

Seed:

$$[0x6A09E667 \oplus (d \cdot 0x9E3779B1) \oplus ((t + 1) \cdot 0x85EBCA77)] \bmod 2^{32}.$$

Dense shell squared radii are stratified uniformly over $[0, 2.2]$ with one deterministic random jitter per bin. Each coordinate gets an independent random sign.

For sparse rows, choose $k \approx 0.25d$ features without replacement and scale their amplitudes so total standardized squared radius matches the sampled shell radius.

S2.7 Model-direction sweep

Collect feature-sized vectors exposed by model parameters and buffers:

- 1-D tensors of size d ;
- rows of tensors whose trailing dimension is d ;
- columns of 2-D tensors with first dimension d and at most 128 columns.

Standardize each candidate direction by current feature scale, normalize it, sort by pre-normalization norm, and greedily keep up to six directions with absolute pairwise cosine no greater than 0.995.

For each direction evaluate 160 equally spaced points over standardized radius

$$[-3.5, 3.5].$$

For each direction, locate the lowest coarse score and refine the adjacent coarse interval with 48 equally spaced points.

S2.8 Model-direction pair probe

Construct model-exposed directions exactly as S2.7, retaining up to four distinct directions.

This probe is used only when the model exposes fewer than two input-facing affine units. Therefore it is active for the one-unit linear model and skipped for the 32-unit MLP.

For every pair of retained directions, evaluate a

$$25 \times 25$$

grid over radius

$$[-3.5, 3.5]^2.$$

For the lowest coarse cell, refine the surrounding coarse interval with a

$$9 \times 9$$

grid.

S2.9 Opaque-corner probe

This probe is only used when the model exposes **zero** input-facing affine units and $d \leq 20$. It is therefore inactive for both model classes reported in this paper.

The frozen Tesla implementation uses the following settings:

- block size 16 features;
- radius 0.8 standardized units;
- enumerate all 2^k sign corners in the first block and, if $d > 16$, the final overlapping block;
- skip the probe when a cheaper pre-opaque margin is already at least 8.0.

S2.10 Between-run spherical probe

Use the last up to eight input mean histories.

Let their coordinate-wise median be c and radial RMS distances

$$r_j = \sqrt{d^{-1} \|M_j - c\|_2^2}.$$

Set

$$R_{\text{RMS}} = 32 \max_j r_j.$$

Let within-run scale be the median over historical feature scales. Skip if

$$R_{\text{RMS}}/s_{\text{within}} < 1.$$

Otherwise generate 768 approximately antipodal random unit directions with deterministic seed

$$[0x243F6A88 \oplus (d \cdot 0x9E3779B1) \oplus ((t + 1) \cdot 0x85EBCA77)] \bmod 2^{32}.$$

Use radial fraction

$$q_i = 0.1 + 0.9 \sqrt{\frac{i + 0.5}{768}},$$

then deterministically permute these fractions.

Probe Euclidean radii

$$R_{\text{RMS}} \sqrt{d} q_i$$

around the historical median center.

S2.11 Affine interaction screen

This probe is active for the MLP and usually inactive for the single-unit linear model because at least two affine boundaries are required.

For every `Linear` layer whose input dimension equals d , each output unit defines an affine boundary

$$w^\top x + b = 0.$$

Express the boundary relative to current feature mean μ and scale σ :

$$w_z = w \odot \sigma,$$

$$a = w^\top \mu + b.$$

The nearest standardized point on that boundary is

$$z_0 = -\frac{a}{\|w_z\|_2^2} w_z.$$

Retain boundaries with

$$\|z_0\|_2 \leq 6.$$

Sort by distance and retain at most 24.

Enumerate exact intersections using pool limits:

```

order 2:  up to 24 units
order 3:  up to 14 units
order 4:  up to 10 units
order 5:  up to  8 units
order 6:  up to  8 units
orders 7-10: up to 10 units
orders 11-12: up to 12 units
orders 13-20:
    nearest order+2 units when total retained units <= 20,
    otherwise nearest order+1 units

```

For a selected set of q boundaries, form matrix A from their standardized normals and vector a from their current offsets. Solve

$$(AA^\top)c = -a,$$

$$z = c^\top A.$$

Reject the intersection if condition number exceeds 10^8 , solve fails, values are non-finite, or

$$\|z\|_2 > 6.$$

Evaluate every valid exact intersection and its input gradient.

Select at most 48 intersections for refinement:

- half from lowest scores;
- the remainder from largest standardized gradient norm;
- fill any duplicates from the next-lowest scores.

Use refinement displacement

$$\epsilon = 0.1.$$

For interaction order $q \leq 6$, enumerate all 2^q signed ± 0.1 offsets along normalized boundary normals.

For $q > 6$, evaluate ± 0.1 along each individual boundary normal plus ± 0.1 along the normalized sum of all selected normals.

The minimum score over exact and refined intersections defines the affine reserve.

S2.12 Disabled generic probes

Two functions present in earlier development versions are **not active in the evaluated frozen Tesla runtime**:

- geometry-support landmark probe;
- older affine-intersection probe.

The frozen Tesla runtime explicitly uses zero for these mechanisms and relies instead on the support/hash machinery and staged affine-combination screen described above.

S3. Exploratory strong-reserve sweep

Before selecting the ten-times median-range comparator, simple reserve controls were evaluated on 40 matched harsh trials.

Combined success rates:

2 x within-run robust score scale:	29 / 40 = 72.5%
5 x within-run robust score scale:	31 / 40 = 77.5%
10 x within-run robust score scale:	32 / 40 = 80.0%
20 x within-run robust score scale:	36 / 40 = 90.0%
Ten-times median-range reserve:	39 / 40 = 97.5%
Full Tesla method on matched harsh set:	40 / 40 = 100.0%

This sweep is explicitly exploratory and is the reason the ten-times median-range comparator cannot be treated as a preregistered comparator.

S4. Statistical formulas

S4.1 Wilson score interval for a binomial proportion

For k successes in n trials and normal quantile z ,

$$\hat{p} = k/n,$$

$$d = 1 + z^2/n,$$

$$c = \frac{\hat{p} + z^2/(2n)}{d},$$

$$h = \frac{z\sqrt{[\hat{p}(1-\hat{p}) + z^2/(4n)]/n}}{d}.$$

Report

$$[c - h, c + h].$$

S4.2 Exact McNemar test for paired binary outcomes

Conditional on $b + c$ discordant pairs under the null of equal paired success probabilities,

$$B \sim \text{Binomial}(b + c, 0.5).$$

Two-sided exact value:

$$p = \min(1, 2 \Pr[B \leq \min(b, c)]).$$

S4.3 Holm correction for multiple planned tests

Sort the two primary raw p -values. Multiply the smallest by 2 and the larger by 1, enforcing monotonic adjusted values and capping at 1.

S4.4 Bootstrap

All paper bootstrap intervals use:

- 20,000 resamples;
- PCG64 with seed 20,260,815;
- complete matched trial pairs for paired effect differences;
- complete model trials for external-label means.

S5. Development-history note

The frozen Tesla implementation was the endpoint of an iterative engineering process before the present manuscript was assembled. The original 13 random-data generator families were part of that broader development environment, which is why this paper does not label the primary 13-family experiment as external validation.

After the Tesla implementation was frozen, eight different challenge generator families were authored and evaluated without changing the Tesla method. Later, during manuscript review, the stronger ten-times median-range comparator was devised after inspecting the behavior of simpler controls. Its subsequent new-seed confirmation is reported transparently as post-selection replication rather than as a pristine confirmatory test.

This chronology is part of the scientific result because it determines which evidence can support which claim.